

Hypoxia time-course RNA-seq pairwise contrast DEGs

Genes highlighted in green had been previously annotated in *Schmidtea mediterranea* within public repositories (Smed_Rink.gtf, or the databases accessed by NCBI BLAST). Hits corresponding to 'hypothetical protein [*Schmidtea mediterranea*]' were not retained. Transcription factor genes highlighted in blue had been previously annotated in *Schmidtea mediterranea* within Neiro *et al.* (2022) AND/OR Pérez-Posada *et al.* (2025).

Genes highlighted in grey had gene annotations based on orthology with *Dugesia japonica* annotations.

For DEGs with no annotations in Dugesiidae, protein sequences predicted by BMKGene BI team from mRNA-seq reads were individually manually inspected. Concordance between the results of BLASTp (*include: Metazoa (taxid:33208)*) and InterProScan analyses of protein sequences was mandated to call family homologs AND/OR orthologs.

0h vs 3h (FC = 1.5, FDR 0.05)

Upregulated genes at 3h (compared to earlier time-point of 0h) ranked from higher magnitude of upregulation:

1. **Smed_histone H1** [h1SMcG0008035] (FDR=1.38142647503189e-16, log2FC=1.36582869723219)
2. **Smed_klf10/11** [h1SMcG0018872] (FDR=0.000900319264954213, log2FC = 0.831172541422798)
3. **Smed_TNF receptor associated factor-2** [h1SMcG0019430] (FDR=0.0042281423545553, log2FC = 0.781950642170977)

Downregulated genes at 3h (compared to earlier time-point of 0h) ranked from higher magnitude of downregulation:

1. **Smed_small HSP protein** [h1SMcG0001680] (FDR=1.20018173138631e-23, log2FC=-2.82455526879826)
2. **Smed_small HSP protein** [h1SMcG0001675] (FDR=3.31360330360461e-21, log2FC=-2.68650721176645)
3. **Smed_small HSP protein** [h1SMcG0001677] (FDR=6.52533836531076e-17, log2FC=-2.67846002166348)
4. **Smed_small HSP protein** [h1SMcG0001676] (FDR=4.29232309793844e-18, log2FC=-2.52377232799697)
5. **Smed_small HSP protein** [h1SMcG0001679] (FDR=9.13831274981075e-18, log2FC=-2.50487356785425)
6. **Smed_small HSP protein** [h1SMcG0001678] (FDR=4.6017477573674e-13, log2FC= -2.41106545979025)
7. **Smed_small HSP protein** [h1SMcG0007183] (FDR=2.84848856235757e-17, log2FC= -2.29193785747494)

8. *Smed_heat shock protein 70-2* [h1SMcG0009604] (FDR= 2.39531061505691e-25, log2FC= -2.06817947569707)
9. *Smed_heat shock protein 70-2* [h1SMcG0009605] (FDR= 1.22235747026866e-36, log2FC= -2.00316148421268)
10. *Smed_heat shock protein 70-2* [h1SMcG0009603] (FDR= 1.1661358804604e-33, log2FC = -1.98112056858094)
11. *Smed_90 kDa heat shock protein* [h1SMcG0005587] (FDR = 6.78207653675728e-13, log2FC = -1.20014445489443)
12. *Smed_EGR-like protein 1* [h1SMcG0008212] (FDR = 6.18135472301488e-06, log2FC = -1.11451944312339)
13. *Smed_Hsp40* [h1SMcG0012561] (FDR = 0.00155399557999026, log2FC = -0.800094503870609)
14. *Smed_stress-induced-phosphoprotein 1* [h1SMcG0013450] (FDR = 0.000151330448197151, log2FC = -0.659757567604799)
15. *Smed_heat shock protein 70-3* [h1SMcG0001183] (FDR = 0.00192458327515966, log2FC = -0.611003927004291)
16. *Smed_serine/threonine-protein kinase Doa-related protein* [h1SMcG0020618] (FDR = 1.75464688966789e-07, log2FC = -0.599574085640558)

0h vs 12h (FC = 1.5, FDR 0.05)

Upregulated genes at 12h (compared to earlier time-point of 0h) ranked from higher magnitude of upregulation, excluding 'ubiquitin' and 'polyubiquitin' hits (h1SMcG0020966, h1SMcG0020968, h1SMcG0020969, h1SMcG0020970, h1SMcG0020975, h1SMcG0020967, h1SMcG0020974 and h1SMcG0020973):

1. *Smed_cellular IAP repeat-containing protein* [h1SMcG0001546] (FDR = 3.58247696879781e-15, log2FC = 2.99891035028143)
2. *Smed_histone H3-like* [h1SMcG0007872] (FDR = 1.27180962521106e-12, log2FC = 2.62169754382767)
3. *Smed_klf10/11* [h1SMcG0018872] (FDR = 1.05380879048782e-57, log2FC = 2.5864125569314)
4. *Smed_histone H1* [h1SMcG0008035] (FDR = 6.95641584140731e-69, log2FC = 2.51063112137179)
5. *Smed_cytochrome P450 2K1-like protein* [h1SMcG0010360] (FDR = 0.00315823475041844, log2FC = 2.13109832035884)
6. *Smed_protein phosphatase 1 regulatory-1* [h1SMcG0010862] (FDR = 1.439478153601e-19, log2FC = 1.7026364144136)
7. *Smed_AC3-33* [h1SMcG0016352] (FDR = 0.000260246060463644, log2FC = 1.6174934743345)
8. *Smed_E3 ubiquitin-protein ligase ARIH1* [h1SMcG0021303] (FDR = 7.2511038862348e-37, log2FC = 1.56533733493046)
9. *Smed_Plac8-E* [h1SMcG0005183] (FDR = 0.00876720604366525, log2FC = 1.49227438636596)
10. *Smed_peptide methionine sulfoxide reductase* [h1SMcG0020033] (FDR = 1.78670638435292e-10, log2FC = 1.36128853710951)

11. *Smed_TNF* receptor-associated factor [h1SMcG0007071] (FDR = 2.71460880215025e-14, log2FC = 1.26019085595799)
12. *Smed_X1.C4.1* [h1SMnG0032112] (FDR = 0.000107856890917652, log2FC = 1.25307395163365)
13. *Smed_TNF* receptor-associated factor 2 [h1SMcG0015086] (FDR = 0.00442531892133795, log2FC = 1.2423025761711)
14. *Smed_histone* 2A-like [h1SMcG0014269] (FDR = 0.00663648828721084, log2FC = 1.23562843232639)
15. *Smed_IAP* domain-containing protein [h1SMcG0007403] (FDR=4.08857186182319e-07, log2FC = 1.22753005785095)
16. *Smed_soma* ferritin [h1SMcG0007370] (FDR = 1.15636098878085e-10, log2FC = 1.2237212162214)
17. *Smed_histone* H4 [h1SMnG0007240] (FDR = 0.0276150359622989, log2FC = 1.19520366201559)
18. *Smed_histone* H4-like [h1SMnG0000572] (FDR = 0.0181673719078538, log2FC = 1.13759800044589)
19. *Smed_ferritin* [h1SMcG0007372] (FDR = 9.51531612747142e-10, log2FC = 1.11847819289312)
20. *Smed_slc20a-2* [h1SMcG0001112] (FDR = 0.00172505298419795, log2FC = 1.09348365780035)
21. *Smed_PIM-1* [h1SMcG0003234] (FDR = 1.04864730044203e-06, log2FC = 1.0531413012311)
22. *Smed_S*-adenosyl-L-methionine-dependent methyltransferase protein [h1SMcG0018161] (FDR = 0.0211693656978791, log2FC = 1.02216225094446)
23. *Smed_protein* phosphatase 1 regulatory subunit [h1SMcG0019740] (FDR = 1.20572697017683e-06, log2FC = 1.00872864569066)
24. *Smed_receptor*-transporting protein [h1SMcG0006507] (FDR = 1.26509654739969e-09, log2FC = 1.00101616452418)
25. *Smed_trans*-L-3-hydroxyproline dehydratase [h1SMcG0015244] (FDR = 2.41858836612481e-05, log2FC = 0.996743020507737)
26. *Smed_sodium*-dependent glucose transporter 1-like [h1SMcG0008847] (FDR = 7.24793760485332e-05, log2FC = 0.987227373013153)
27. *Smed_folliculin*-interacting protein 2-like [h1SMcG0006386] (FDR = 1.27791037306851e-09, log2FC = 0.968043261519269)
28. *Smed_thialysine* N-epsilon-acetyltransferase-like [h1SMcG0006077] (FDR = 0.00404688874552385, log2FC = 0.948648776432487)
29. *Smed_pantothenate* kinase protein [h1SMcG0016781] (FDR = 4.99850900567525e-10, log2FC = 0.940852614258702)
30. *Smed_coiled-coil* domain containing protein 103-like [h1SMcG0002927] (FDR = 0.0175986609039387, log2FC = 0.915456695065911)
31. *Smed_AFG1*-like ATPase [h1SMcG0022995] (FDR = 0.0495940621284536, log2FC = 0.880048322473888)
32. *Smed_pyruvate* dehydrogenase E1 subunit beta [h1SMnG0021891] (FDR = 0.0228968573904775, log2FC = 0.87233100026623)
33. *Smed_steroid* 17-alpha-hydroxylase/17,20 lyase-like [h1SMcG0005150] (FDR = 1.73742460246961e-12, log2FC = 0.86524341808731)

34. **Smed_X1.A4.3** [h1SMnG0020921] (FDR = 5.52479443891034e-06, log2FC = 0.862737853340981)
35. *Smed_D-3-phosphoglycerate dehydrogenase-like* [h1SMcG0017641] (FDR = 7.93893844514201e-08, log2FC = 0.860753646715536)
36. **Smed_lactB** [h1SMcG0008196] (FDR = 0.000413521185109529, log2FC = 0.854764853305172)
37. *Smed_OSTM1* [h1SMcG0015181] (FDR = 0.00530389356369653, log2FC = 0.84265078567945)
38. **Smed_epidermal growth factor-3** [h1SMnG0019034] (FDR = 0.00748871907071208, log2FC = 0.83001386609272)
39. **Smed_tlx** [h1SMcG0020740] (FDR = 0.0416128172866063, log2FC = 0.789040583421669)
40. *Smed_alpha-tocopherol transfer protein* [h1SMcG0022485] (FDR = 0.0108244547013636, log2FC = 0.788803830648314)
41. *Smed_protein phosphatase 1 regulatory subunit 3* [h1SMcG0004471] (FDR = 1.27791037306851e-09, log2FC = 0.78331973517605)
42. *Smed_histone 2A-like* [h1SMcG0014269] (FDR = 0.0189301167550125, log2FC = 0.753112090148396)
43. *Smed_CXXC motif containing zinc binding protein* [h1SMcG0003797] (FDR = 0.0438603919162878, log2FC = 0.751143596710308)
44. *Smed_basic-leucine zipper (bZIP) domain transcription factor* [h1SMnG0004780] (FDR = 4.60208209750377e-05, log2FC = 0.750783137993124)
45. *Smed_RNA-binding protein* [h1SMcG0004822] (FDR = 0.00442531892133795, log2FC = 0.746439149897278)
46. **Smed_caveolin-1** [h1SMcG0007790] (FDR = 0.0250338708637254, log2FC = 0.739115759250286)
47. *Smed_transmembrane protein 231-like* [h1SMcG0016540] (FDR = 0.00442700870325177, log2FC = 0.728405484043073)
48. *Smed_methylenetetrahydrofolate reductase (NADPH)-like* [h1SMcG0005986] (FDR = 2.25315112077937e-08, log2FC = 0.725963003873156)
49. *Smed_sideroflexin-1* [h1SMcG0006070] (FDR = 7.54748968940734e-06, log2FC = 0.725506830739907)
50. *Smed_haloacid dehalogenase-like hydrolase domain-containing protein 3* [h1SMcG0017099] (FDR = 0.000380557043299876, log2FC = 0.723867501684938)
51. *Smed_E3 ubiquitin ligase RBR family protein* [h1SMcG0004408] (FDR = 3.8688529692577e-07, log2FC = 0.72362613218625)
52. *Smed_lipopolysaccharide-induced tumour necrosis factor-alpha factor* [h1SMcG0015134] (FDR = 0.0288684169364634, log2FC = 0.702822052427758)
53. *Smed_mpv17-like protein* [h1SMcG0012089] (FDR = 0.00121506896391192, log2FC = 0.694872019788812)
54. *Smed_B9 domain-containing protein 2-like* [h1SMcG0005869] (FDR = 0.0463039934802364, log2FC = 0.692130183106459)
55. **Smed_TNF receptor associated factor-3** [h1SMcG0000388] (FDR = 0.0119296043337979, log2FC = 0.68888305034398)
56. **Smed_blitzschnell family protein** [h1SMcG0017122] (FDR = 0.0346203581680332, log2FC = 0.688024474491003)
57. *Smed_FGFR1 oncogene partner/LisH domain-containing protein* [h1SMcG0008329] (FDR = 0.0156664288385699, log2FC = 0.684149334963114)

58. *Smed_casein kinase 1-like* [h1SMcG0009853] (FDR = 0.0255293010967844, log2FC = 0.682271145495367)
59. *Smed_JNK1/MAPK8-associated membrane protein-like* [h1SMcG0003989] (FDR = 0.0188824904746667, log2FC = 0.675442121571059)
60. *Smed_universal stress protein* [h1SMcG0009643] (FDR = 0.000207511516038525, log2FC = 0.667448557348182)
61. *Smed_origin recognition complex subunit 5-like* [h1SMcG0004422] (FDR = 0.0468079825279029, log2FC = 0.647059522609738)
62. *Smed_coiled-coil domain containing protein 42* [h1SMcG0005935] (FDR = 0.0371614376071435, log2FC = 0.637203165286036)
63. *Smed_EF-hand calcium-binding domain profile protein* [h1SMnG0006596] (FDR = 0.0493913679682929, log2FC = 0.635382522415161)
64. *Smed_elf3* [h1SMcG0016104] (FDR = 0.00663648828721084, log2FC = 0.630082787292046)
65. *Smed_monoglyceride lipase-like* [h1SMcG0018156] (FDR = 0.000962836530441079, log2FC = 0.630041004649318)
66. *Smed_N(6)-adenine-specific methyltransferase METTL4-like* [h1SMcG0014536] (FDR = 0.0161783447615858, log2FC = 0.629656655501488)
67. *Smed_E3 ubiquitin-protein ligase RNF170-like* [h1SMcG0011093] (FDR = 0.0286501956854926, log2FC = 0.623804784818577)
68. *Smed_noggin-like protein 3* [h1SMcG0003617] (FDR = 0.00741404121515056, log2FC = 0.619106148794592)
69. *Smed_acidic fibroblast growth factor intracellular-binding protein* [h1SMcG0021096] (FDR = 0.0498550732620391, log2FC = 0.61717571046452)
70. *Smed_pantothenate kinase protein* [h1SMcG0016782] (FDR = 7.47608789265894e-06, log2FC = 0.61594769956299)
71. *Smed_V-type proton ATPase subunit G-like* [h1SMnG0017027] (FDR = 0.0156664288385699, log2FC = 0.615256472653744)
72. *Smed_P450 2B-like protein* [h1SMcG0009079] (FDR = 0.0016840690073405, log2FC = 0.614196995301107)
73. *Smed_NB.21.11e* [h1SMcG0015271] (FDR = 0.00644221362301043, log2FC = 0.608563160458727)
74. *Smed_inositol phosphoceramide mannosyltransferase 2-like* [h1SMcG0002184] (FDR = 0.0189723196767602, log2FC = 0.604605920116315)
75. *Smed_phospholipase A2-like* [h1SMcG0017624] (FDR = 0.000307904754259071, log2FC = 0.59152647534885)
76. *Smed_polynucleotide 5'-hydroxyl-kinase nol9* [h1SMcG0011881] (FDR = 0.0387778170969909, log2FC = 0.586923768396373)
77. *Smed_epidermal growth factor-1* [h1SMcG0022114] (FDR = 0.00899749818151532, log2FC = 0.585433916490715)

Downregulated genes at 12h (compared to earlier time-point of 0h) ranked from higher magnitude of downregulation:

1. *Smed_small HSP protein* [h1SMcG0001680] (FDR = 3.12411335036831e-09, log2FC = -2.53046938075355)

2. *Smed_small HSP protein* [h1SMcG0001677] (FDR = 6.71366351758048e-07, log2FC = -2.50095001583133)
3. *Smed_heat shock protein 70-2* [h1SMcG0009604] (FDR = 2.50521528802747e-57, log2FC = -2.48907254360058)
4. *Smed_heat shock protein 70-2* [h1SMcG0009605] (FDR = 2.34113185346732e-57, log2FC = -2.45069476404026)
5. *Smed_heat shock protein 70-2* [h1SMcG0009603] (FDR = 1.77166461407254e-56, log2FC = -2.44767181735275)
6. *Smed_small HSP protein* [h1SMcG0001675] (FDR = 1.81852351156331e-07, log2FC = -2.44368048592877)
7. *Smed_EGR-like protein 1* [h1SMcG0008212] (FDR = 2.67943780692165e-35, log2FC = -2.3815585389635)
8. *Smed_early growth response-3* [h1SMcG0008213] (FDR = 3.54935819070462e-15, log2FC = -2.35004994599118)
9. *Smed_small HSP protein* [h1SMcG0007183] (FDR = 8.88064196219861e-22, log2FC = -2.33773119055451)
10. *Smed_small HSP protein* [h1SMcG0001679] (FDR = 3.4316597222534e-18, log2FC = -2.28927587965816)
11. *Smed_small HSP protein* [h1SMcG0001676] (FDR = 4.47276243434447e-06, log2FC = -2.2789831245795)
12. *Smed_small HSP protein* [h1SMcG0001678] (FDR = 1.15636098878085e-10, log2FC = -2.01517642097804)
13. *Smed_90 kDa heat shock protein* [h1SMcG0005587] (FDR = 6.49351083048253e-66, log2FC = -1.91444646982037)
14. *Smed_ankyrin repeat domain-containing protein* [h1SMcG0023020] (FDR = 4.08375434759559e-05, log2FC = -1.79985141730972)
15. *Smed_ankyrin repeat domain-containing protein* [h1SMnG0032253] (FDR = 0.00663648828721084, log2FC = -1.78076581498412)
16. *Smed_small HSP protein* [h1SMcG0007183] (FDR = 0.0267882842597759, log2FC = -1.39465705360132)
17. *Smed_small subunit processome component 20 homolog* [h1SMcG0004349] (FDR = 2.60872191600385e-06, log2FC = -1.36269409342603)
18. *Smed_transposon Ty3-G Gag-Pol polyprotein* [h1SMcG0017868] (FDR = 8.08912534599225e-05, log2FC = -1.29662179326601)
19. *Smed_E3 ubiquitin-protein ligase* [h1SMcG0022920] (FDR = 0.0194353769250479, log2FC = -1.29154148029926)
20. *Smed_slc18a-8* [h1SMcG0010030] (FDR = 0.00221074304481024, log2FC = -1.23200460501081)
21. *Smed_plexin B-5* [h1SMcG0022656] (FDR = 0.0145576273051484, log2FC = -1.14378672623933)
22. *Smed_GCR022* [h1SMcG0004902] (FDR = 0.0167960805967277, log2FC = -1.07928016465078)
23. *Smed_pre-mRNA-splicing factor 8* [h1SMcG0006598] (FDR = 3.50364244160221e-05, log2FC = -0.926746297923045)
24. *Smed_Hsp40* [h1SMcG0012561] (FDR = 3.91116502755277e-06, log2FC = -0.923120488102438)
25. *Smed_unc-80-like* [h1SMcG0019733] (FDR = 0.00384013662712213, log2FC = -0.916173371421335)

26. *Smed_tubulin/FtsZ, GTPase domain protein* [h1SMnG0016149] (FDR = 0.00901072925446809, log2FC = -0.855869650965162)
27. *Smed_Lost PHD-fingers of trithorax-related* [h1SMcG0016489] (FDR = 0.00384013662712213, log2FC = -0.851259292901032)
28. *Smed_mediator of RNA polymerase II transcription subunit-12 like* [h1SMcG0008216] (FDR = 0.0179690630625701, log2FC = -0.84402663029616)
29. *Smed_slc4a-4* [h1SMcG0014894] (FDR = 5.31810942879076e-07, log2FC = -0.833980348605392)
30. *Smed_alkaline phosphatase-like* [h1SMcG0002566] (FDR = 0.0176886995026913, log2FC = -0.833148309394761)
31. *Smed_PH-like domain-containing protein* [h1SMcG0005736] (FDR = 9.08325989971901e-07, log2FC = -0.825658987620468)
32. *Smed_ectopic P granules protein 5 homolog* [h1SMcG0003834] (FDR = 0.0127869706265884, log2FC = -0.779988729385078)
33. *Smed_heat shock protein 70-3* [h1SMcG0001183] (FDR = 1.02040371943068e-08, log2FC = -0.76924612397311)
34. *Smed_serine-protein kinase ATM-like* [h1SMcG0001804] (FDR = 0.028979562054338, log2FC = -0.759244280424226)
35. *Smed_BAG domain-containing protein* [h1SMcG0004717] (FDR = 0.00478373080656585, log2FC = -0.747847378231941)
36. *Smed_PKD1L-3* [h1SMcG0019279] (FDR = 0.00876720604366525, log2FC = -0.743505843442354)
37. *Smed_stress-induced-phosphoprotein 1* [h1SMcG0013450] (FDR = 2.24385363897913e-07, log2FC = -0.74335313412472)
38. *Smed_activator of 90 kDa heat shock protein ATPase* [h1SMcG0017018] (FDR = 8.79225728987823e-07, log2FC = -0.734036324022459)
39. *Smed_C2H2-type zinc finger domain-containing protein* [h1SMcG0006557] (FDR = 0.00161654246435734, log2FC = -0.733944765299401)
40. *Smed_filamin-A* [h1SMcG0013445] (FDR = 0.0275291532860057, log2FC = -0.699358754554205)
41. *Smed_PKD1L-1* [h1SMcG0019281] (FDR = 0.0399061115592269, log2FC = -0.695705690251486)
42. *Smed_rpb1* [h1SMcG0010290] (FDR = 0.0389601690556611, log2FC = -0.693675658037667)
43. *Smed_ABCC1-like* [h1SMcG0012648] (FDR = 0.028979562054338, log2FC = -0.691806770494205)
44. *Smed_teneurin-m-like* [h1SMcG0021003] (FDR = 0.0319450986776509, log2FC = -0.687039091364777)
45. *Smed_pre-mRNA cleavage complex 2-like* [h1SMcG0001671] (FDR = 0.0288684169364634, log2FC = -0.680700908900397)
46. *Smed_T-complex protein 1 subunit beta* [h1SMcG0021008] (FDR = 0.00876720604366525, log2FC = -0.678646635035233)
47. *Smed_integrator complex subunit 1* [h1SMcG0020920] (FDR = 0.0129169835291342, log2FC = -0.676735013656826)
48. *Smed_voltage operated calcium channel Cav1B* [h1SMcG0000076] (FDR = 0.0161827397108498, log2FC = -0.673579662548661)
49. *Smed_ADAMTS domain-containing protein* [h1SMcG0018115] (FDR = 0.00741404121515056, log2FC = -0.670818272120542)

50. *Smed_SEZ6/CSMD family protein* [h1SMcG0003529] (FDR = 0.00741404121515056, log2FC = -0.6653300877141)
51. *Smed_serine/threonine-protein kinase Doa-related protein* [h1SMcG0020618] (FDR = 2.57591297120949e-09, log2FC = -0.663824244402757)
52. *Smed_CAP domain protein* [h1SMcG0020022] (FDR = 0.0188824904746667, log2FC = -0.661377516858219)
53. *Smed_cadherin domain-containing protein* [h1SMcG0022497] (FDR = 0.0195280870689421, log2FC = -0.658078757833118)
54. *Smed_transcription elongation regulator 1* [h1SMcG0014961] (FDR = 0.0114374105498493, log2FC = -0.640879015948489)
55. *Smed_DPB11-related* [h1SMcG0014842] (FDR = 0.0224751600587327, log2FC = -0.638661326210767)
56. *Smed_titin-like protein* [h1SMcG0014354] (FDR = 0.0289089950547431, log2FC = -0.637877727449401)
57. *Smed_claspin* [h1SMcG0016426] (FDR = 0.0173450181456266, log2FC = -0.635362418616671)
58. *Smed_tyrosine-protein phosphatase Lar-like* [h1SMcG0013355] (FDR = 0.0280511857070909, log2FC = -0.633505270695659)
59. *Smed_rpb1* [h1SMcG0020592] (FDR = 0.00255603639590523, log2FC = -0.629542919132948)
60. *Smed_TFIIIB component B* [h1SMcG0022577] (FDR = 0.0412483465905669, log2FC = -0.623676161381771)
61. *Smed_ankyrin repeat domain-containing protein* [h1SMcG0005883] (FDR = 0.0400483272134308, log2FC = -0.618057639078038)
62. *Smed_cytoplasmic FMR1-interacting protein-related* [h1SMcG0022633] (FDR = 7.54423053497729e-07, log2FC = -0.61509960404215)
63. *Smed_receptor-type tyrosine-protein phosphatase eta-like* [h1SMcG0004832] (FDR = 0.0456406026096304, log2FC = -0.610132224023552)
64. *Smed_CDON/DCC-like adhesion receptor* [h1SMcG0016404] (FDR = 0.0099294996263180, log2FC = -0.609323374789803)
65. *Smed_fibrillin-like protein* [h1SMcG0011212] (FDR = 0.0149048001244054, log2FC = -0.60290582165356)
66. *Smed_Kunitz-type serine protease inhibitor* [h1SMcG0000534] (FDR = 0.0494631257185481, log2FC = -0.601809521826973)
67. *Smed_cytochrome P450 superfamily protein* [h1SMcG0010229] (FDR = 0.0193912307612636, log2FC = -0.587351281077048)

0h vs 24h (FDR = 0.05, FC = 1.5)

Upregulated genes at 24h (compared to earlier time-point of 0h) ranked from higher magnitude of upregulation, excluding 'ubiquitin' and 'polyubiquitin' hits (h1SMcG0020966, h1SMcG0020968, h1SMcG0020974, h1SMcG0020969, h1SMcG0020967, h1SMcG0020975, h1SMcG0020971, h1SMcG0020970 and h1SMcG0020973):

1. *Smed_cellular IAP repeat-containing protein* [h1SMcG0001546] (FDR = 5.15762571099979e-25, log2FC = 4.67718734694103)

2. **Smed_TNF receptor-associated factor 2** [h1SMcG0015086] (FDR = 1.12086643978028e-20, log2FC = 2.70820804010197)
3. *Smed_E3 ubiquitin-protein ligase ARIH1* [h1SMcG0021303] (FDR = 1.02005309076718e-98, log2FC = 2.70749665027098)
4. *Smed_soma ferritin* [h1SMcG0007370] (FDR = 2.06653165235692e-51, log2FC = 2.66961813425779)
5. *Smed_histone H3-like* [h1SMcG0007872] (FDR = 4.29714606339831e-11, log2FC = 2.56003936475073)
6. **Smed_histone H1** [h1SMcG0008035] (FDR = 4.13320477040757e-48, log2FC = 2.47374985390147)
7. **Smed_klf10/11** [h1SMcG0018872] (FDR = 2.67848619827999e-53, log2FC = 2.47241732290004)
8. *Smed_ferritin* [h1SMcG0007372] (FDR = 2.26098489730019e-35, log2FC = 2.33010371639929)
9. *Smed_peptide methionine sulfoxide reductase* [h1SMcG0020033] (FDR = 1.86552829467131e-34, log2FC = 2.1870831756916)
10. *Smed_pantothenate kinase protein* [h1SMcG0016781] (FDR = 8.01429691585929e-39, log2FC = 2.13504657314262)
11. *Smed_E3 ubiquitin ligase RBR family protein* [h1SMcG0004408] (FDR = 7.32923840522557e-73, log2FC = 2.01474059101616)
12. *Smed_TNF receptor-associated factor* [h1SMcG0007071] (FDR = 1.595491881494e-30, log2FC = 1.72814792933042)
13. *Smed_cytochrome P450 2K1-like protein* [h1SMcG0010360] (FDR = 2.91445151491843e-07, log2FC = 1.72790084233783)
14. **Smed_slc20a-2** [h1SMcG0001112] (FDR = 3.90605307211214e-07, log2FC = 1.57484155679645)
15. *Smed_pantothenate kinase protein* [h1SMcG0016782] (FDR = 3.16866390958565e-26, log2FC = 1.45663577328193)
16. *Smed_sulfotransferase protein* [h1SMcG0008457] (FDR = 0.000870153434495039, log2FC = 1.42160788057888)
17. *Smed_receptor-transporting protein* [h1SMcG0006507] (FDR = 1.07437975809788e-09, log2FC = 1.31131446427535)
18. *Smed_consortin-like* [h1SMcG0022227] (FDR = 0.0154613181636795, log2FC = 1.27332633438307)
19. *Smed_trans-L-3-hydroxyproline dehydratase* [h1SMcG0015244] (FDR = 1.6739565518057e-07, log2FC = 1.27125448286135)
20. *Smed_protein phosphatase 1 regulatory subunit* [h1SMcG0019740] (FDR = 5.72909830831485e-09, log2FC = 1.23897560451165)
21. *Smed_TNF receptor-associated factor* [h1SMcG0022911] (FDR = 0.000187356957952556, log2FC = 1.23834484285284)
22. **Smed_epidermal growth factor-1** [h1SMcG0022114] (FDR = 9.87669550857311e-12, log2FC = 1.21863722462662)
23. *Smed_Obg-like GTPase* [h1SMcG0006933] (FDR = 0.0403207697862251, log2FC = 1.21738338962018)
24. *Smed_AFG1-like ATPase* [h1SMcG0022995] (FDR = 0.00106270683870888, log2FC = 1.20933102884505)

25. *Smed_universal stress protein* [h1SMcG0009643] (FDR = 2.50897731818373e-12, log2FC = 1.15839109563416)
26. *Smed_family A G protein-coupled receptor* [h1SMcG0009373] (FDR = 0.0270938737827564, log2FC = 1.1105133748283)
27. *Smed_ankyrin repeat domain-containing protein* [h1SMnG0029278] (FDR = 0.0335087082983479, log2FC = 1.090169490003)
28. *Smed_RNA-binding protein 48* [h1SMcG0005509] (FDR = 0.0278539099714641, log2FC = 1.08255437613308)
29. *Smed_basic-leucine zipper (bZIP) domain transcription factor* [h1SMnG0004780] (FDR = 0.000108506289677869, log2FC = 1.04707268656742)
30. *Smed_PIM-1* [h1SMcG0003234] (FDR = 0.0122817369479367, log2FC = 1.02702592791757)
31. *Smed_elf3* [h1SMcG0016104] (FDR = 0.0096187852271878, log2FC = 1.00467622971804)
32. *Smed_MAPK phosphatase* [h1SMcG0019962] (FDR = 1.5781819882242e-11, log2FC = 0.998218301994907)
33. *Smed_major facilitator superfamily domain-containing protein 6-like* [h1SMcG0014483] (FDR = 2.08028269298557e-05, log2FC = 0.971530774610203)
34. *Smed_protein phosphatase 1 regulatory subunit 3-like* [h1SMcG0004471] (FDR = 2.00338917818561e-13, log2FC = 0.969735774123192)
35. *Smed_monoglyceride lipase-like* [h1SMcG0018156] (FDR = 1.04072378671335e-05, log2FC = 0.93694134203998)
36. *Smed_5'-AMP-activated protein kinase, gamma regulatory subunit* [h1SMcG0005059] (FDR = 2.6011727678569e-10, log2FC = 0.936172542604209)
37. *Smed_protein phosphatase 1 regulatory-1* [h1SMcG0010862] (FDR = 7.99433936962127e-06, log2FC = 0.933140398989467)
38. *Smed_adiponectin receptor protein-like* [h1SMcG0013451] (FDR = 1.24385223550662e-11, log2FC = 0.896022928864744)
39. *Smed_monocarboxylate transporter protein* [h1SMcG0016325] (FDR = 0.00465331868716048, log2FC = 0.895491948391842)
40. *Smed_ferritin-like protein* [h1SMcG0011235] (FDR = 2.85455734072406e-05, log2FC = 0.879672391443027)
41. *Smed_phospholipase A2-like* [h1SMcG0017624] (FDR = 2.44060074292245e-07, log2FC = 0.871389856054089)
42. *Smed_cytochrome P450 2B-like protein* [>SMEST072744002.1.p1 type:complete len:507 gc:universal SMEST072744002.1:392-1912(+)] (FDR = 1.07666792904076e-06, log2FC = 0.871177186506046)
43. *Smed_globin-like superfamily protein* [h1SMcG0015726] (FDR = 0.0246816643899161, log2FC = 0.849689337825815)
44. *Smed_OSTM1* [h1SMcG0015181] (FDR = 0.0141745627538948, log2FC = 0.847510145576924)
45. *Smed_Ras-family small GTPase* [h1SMcG0015679] (FDR = 0.00361499416653575, log2FC = 0.826928332077971)
46. *Smed_methylenetetrahydrofolate reductase (NADPH)-like* [h1SMcG0005986] (FDR = 2.71650960130394e-07, log2FC = 0.788475672309611)
47. *Smed_steroid 17-alpha-hydroxylase/17,20 lyase-like* [h1SMcG0005150] (FDR = 1.6739565518057e-07, log2FC = 0.781027324101094)

48. **Smed_slc24a-5** [h1SMnG0034353] (FDR = 0.0271781537941673, log2FC = 0.776917335631588)
49. **Smed_P450 2B-like protein** [h1SMcG0009079] (FDR = 6.36665699499414e-05, log2FC = 0.772912844357347)
50. **Smed_1-Jun** [h1SMcG0021300] (FDR = 6.75335123625404e-06, log2FC = 0.759571550282109)
51. **Smed_S-adenosyl-L-methionine-dependent methyltransferase protein** [h1SMcG0000398] (FDR = 0.0218634187876964, log2FC = 0.752445187838223)
52. **Smed_D-3-phosphoglycerate dehydrogenase-like** [h1SMcG0017641] (FDR = 0.000258901186954124, log2FC = 0.751240734324512)
53. **Smed_slc25a-29** [h1SMcG0020425] (FDR = 9.23426924236948e-06, log2FC = 0.746877688050824)
54. **Smed_2-aminoethanethiol dioxygenase** [h1SMcG0005435] (FDR = 4.94103163656047e-06, log2FC = 0.744199095721098)
55. **Smed_mpv17-like protein** [h1SMcG0021924] (FDR = 0.00106879341058351, log2FC = 0.742806587537975)
56. **Smed_globin-like superfamily protein** [h1SMcG0003105] (FDR = 0.00686824537008717, log2FC = 0.709754276719074)
57. **Smed_membrane-associated progesterone receptor component-related** [h1SMcG0016165] (FDR = 0.0129930505376985, log2FC = 0.700864049662696)
58. **Smed_pseudo-PARP domain-containing protein** [h1SMcG0001523] (FDR = 6.7788162903354e-05, log2FC = 0.693638546891132)
59. **Smed_NAD(P)-dependent oxidoreductase** [h1SMcG0007023] (FDR = 0.0004042304692202287, log2FC = 0.6854383188696)
60. **Smed_ABCD4-like ABC transporter** [h1SMcG0002437] (FDR = 0.000876100663574507, log2FC = 0.67609352119454)
61. **Smed_folliculin-interacting protein 2-like** [h1SMcG0006386] (FDR = 0.00451923814855616, log2FC = 0.660646371282663)
62. **Smed_dynamin-1-like** [h1SMcG0013705] (FDR = 0.00355016809915237, log2FC = 0.659319604368408)
63. **Smed_fatty acid-binding protein** [h1SMcG0000553] (FDR = 0.0299939012408775, log2FC = 0.650602245444155)
64. **Smed_FAD-dependent oxidoreductase domain-containing protein 1-like** [h1SMcG0004813] (FDR = 2.47621850181648e-05, log2FC = 0.646169390449471)
65. **Smed_phosphoserine aminotransferase-like** [h1SMcG0020665] (FDR = 8.5095586787906e-06, log2FC = 0.64142855773635)
66. **Smed_caspase-3C** [h1SMcG0020890] (FDR = 0.0218634187876964, log2FC = 0.627214564147399)
67. **Smed_RNA polymerase III subunit RPC3-like** [h1SMcG0005276] (FDR = 0.00287408712608569, log2FC = 0.624470433394357)
68. **Smed_slc29a-1** [h1SMcG0006403] (FDR = 0.00111683359469384, log2FC = 0.61724856438914)
69. **Smed_sulfotransferase protein** [h1SMcG0020315] (FDR = 0.00136974022046967, log2FC = 0.605384692357915)

Downregulated genes at 24h (compared to earlier time-point of 0h) ranked from higher magnitude of downregulation:

1. *Smed_P-type ATPase* [h1SMcG0023126] (FDR = 9.57107546753765e-14, log2FC = -6.28249656793293)
2. *Smed_early growth response-3* [h1SMcG0008213] (FDR = 5.9699953240264e-12, log2FC = -2.73146845437671)
3. *Smed_TBC1 domain-containing Rab GTPase-activating protein* [h1SMcG0023066] (FDR = 0.00572044798044765, log2FC = -2.63876401215714)
4. *Smed_EGR-like protein 1* [h1SMcG0008212] (FDR = 5.32119722912003e-27, log2FC = -2.62966637102936)
5. *Smed_ankyrin repeat domain-containing protein* [h1SMnG0032253] (FDR = 0.00312488214941413, log2FC = -2.05096931149418)
6. *Smed_heat shock protein 70-2* [h1SMcG0009604] (FDR= 5.52347624185791e-39, log2FC= -2.03089738561356)
7. *Smed_small HSP protein* [h1SMcG0001680] (FDR= 7.81454930224796e-10, log2FC = -2.01253465369585)
8. *Smed_heat shock protein 70-2* [h1SMcG0009603] (FDR= 3.91727761726611e-32, log2FC = -1.98282482422157)
9. *Smed_small HSP protein* [h1SMcG0001677] (FDR= 7.20860720385898e-08, log2FC= -1.93986274240329)
10. *Smed_heat shock protein 70-2* [h1SMcG0009605] (FDR= 1.595491881494e-30, log2FC= -1.90713508877028)
11. *Smed_small HSP protein* [h1SMcG0001675] (FDR = 1.25310920839094e-08, log2FC = -1.8894644698275)
12. *Smed_small HSP protein* [h1SMcG0001679] (FDR= 2.29754322984876e-07, log2FC= -1.86150109615355)
13. *Smed_small HSP protein* [h1SMcG0001676] (FDR = 4.88383314908123e-07, log2FC = -1.84189509120105)
14. *Smed_small HSP protein* [h1SMcG0007183] (FDR= 2.65982520467172e-10, log2FC= -1.81703050797017)
15. *Smed_90 kDa heat shock protein* [h1SMcG0005587] (FDR = 2.21974182900885e-33, log2FC = -1.66831438745081)
16. *Smed_small HSP protein* [h1SMcG0001678] (FDR= 7.66635295741904e-05, log2FC= -1.61413817095601)
17. *Smed_peptidase M8 family metalloprotease* [h1SMcG0008332] (FDR = 0.000777653470572521, log2FC = -1.38723911413941)
18. *Smed_PH-like domain-containing protein* [h1SMcG0005736] (FDR = 3.10040609018044e-09, log2FC = -1.26772165311974)
19. *Smed_cytochrome P450 superfamily protein* [h1SMcG0010229] (FDR = 3.62692059797148e-06, log2FC = -1.08854509185721)
20. *Smed_GCR020* [h1SMcG0021033] (FDR = 0.0383559623714035, log2FC = -1.06289677615819)
21. *Smed_alkaline phosphatase-like* [h1SMcG0002566] (FDR = 0.00217018335899168, log2FC = -1.05812829103275)
22. *Smed_Kunitz-type serine protease inhibitor* [h1SMcG0000534] (FDR = 0.00447618839382525, log2FC = -1.04839481115697)

23. *Smed_short-chain dehydrogenase/reductase (SDR) family protein* [h1SMcG0010682] (FDR = 0.0262321823839263, log2FC = -1.02621830991063)
24. *Smed_beta-1,3-N-acetylglucosaminyltransferase-like* [h1SMcG0022263] (FDR = 9.18984053171771e-08, log2FC = -0.969939624864481)
25. *Smed_slc7a-5* [h1SMcG0021830] (FDR = 0.00387508524459369, log2FC = -0.959161610145726)
26. *Smed_sulfotransferase protein* [h1SMcG0020062] (FDR = 0.0398591334366317, log2FC = -0.921366875916586)
27. *Smed_Hsp40* [h1SMcG0012561] (FDR = 0.000697206749176895, log2FC = -0.909610442479301)
28. *Smed_ADAMTS domain-containing protein* [h1SMcG0018115] (FDR = 0.000161963132877943, log2FC = -0.872020057877953)
29. *Smed_T-complex protein 1 subunit beta* [h1SMcG0021008] (FDR = 0.00251993754614677, log2FC = -0.826590213739906)
30. *Smed_short-chain dehydrogenase/reductase (SDR) family protein* [h1SMcG0010684] (FDR = 0.022298971076354, log2FC = -0.822948827336753)
31. *Smed_pre-mRNA cleavage complex 2-like* [h1SMcG0001671] (FDR = 0.00308742233845324, log2FC = -0.802030194237209)
32. *Smed_stress-induced-phosphoprotein 1* [h1SMcG0013450] (FDR = 0.000375696535492958, log2FC = -0.729317696902997)
33. *Smed_slc4a-4* [h1SMcG0014894] (FDR = 1.22423064585461e-05, log2FC = -0.7133629303195)
34. *Smed_activator of 90 kDa heat shock protein ATPase* [h1SMcG0017018] (FDR = 0.000222336860096008, log2FC = -0.656937883691813)
35. *Smed_class B secretin-like GPCR* [h1SMcG0005956] (FDR = 7.99810358762492e-05, log2FC = -0.656723416354398)
36. *Smed_serine/threonine-protein phosphatase 1 regulatory subunit 10-like* [h1SMcG0014197] (FDR = 3.93191507963862e-06, log2FC = -0.656371914848266)
37. *Smed_WD40 repeat-containing G protein beta-like protein* [h1SMcG0007365] (FDR = 0.0331224300297319, log2FC = -0.650434447457112)
38. *Smed_diacylglycerol kinase protein* [h1SMcG0000617] (FDR = 0.041246159797509, log2FC = -0.643000427543618)
39. *Smed_serine/threonine-protein kinase Doa-related protein* [h1SMcG0020618] (FDR = 3.12615217737746e-05, log2FC = -0.638511015564955)
40. *Smed_O-acyltransferase (WSD1-like) family protein* [h1SMcG0015166] (FDR = 0.00411005525997754, log2FC = -0.627225177934219)
41. *Smed_BCKHDA-like protein* [h1SMcG0015276] (FDR = 0.00820310923666195, log2FC = -0.624231396977168)
42. *Smed_heat shock protein 70-3* [h1SMcG0001183] (FDR = 0.00418101873077047, log2FC = -0.617628627473148)
43. *Smed_RNA-binding protein* [h1SMcG0010534] (FDR = 0.00781956651197401, log2FC = -0.608679507257528)
44. *Smed_ATP-dependent RNA helicase* [h1SMcG0009222] (FDR = 0.0455214571425619, log2FC = -0.606188358647598)
45. *Smed_TNF receptor associated factor-2* [h1SMnG0015311] (FDR = 0.0107759056584525, log2FC = -0.590978665344013)
46. *Smed_receptor-type tyrosine-protein phosphatase* [h1SMcG0001587] (FDR = 0.0323270637633812, log2FC = -0.588144052425185)

47. *Smed_phosphatidylserine synthase protein* [h1SMnG0020573] (FDR = 0.0314314169994216, log2FC = -0.586763439711929)

0h vs 36h (FDR = 0.05, FC = 1.5)

Upregulated genes at 36h (compared to earlier pre-hypoxia normoxia time-point of 0h), excluding 'ubiquitin' and 'polyubiquitin' hits (h1SMcG0020968, h1SMcG0020966, h1SMcG0020967, h1SMcG0020975, h1SMcG0020969, h1SMcG0020970, h1SMcG0020974, h1SMcG0020956, h1SMcG0016320) ranked from higher magnitude of upregulation:

1. *Smed_sulfotransferase protein* [h1SMcG0008457] (FDR = 1.00532717377745e-13, log2FC = 2.21464598973724)
2. *Smed_X1.C4.1* [h1SMnG0032112] (FDR = 2.64592531668576e-08, log2FC = 1.47281168782236)
3. *Smed_COP9 signalosome complex subunit 8-like protein* [h1SMcG0013441] (FDR = 0.0417304561564794, log2FC = 1.37955990783769)
4. *Smed_ston* [h1SMcG0006168] (FDR = 2.45819872879459e-05, log2FC = 1.34291200560893)
5. *Smed_ferritin* [h1SMcG0007372] (FDR = 1.42301017570569e-16, log2FC = 1.29178893395334)
6. *Smed_soma ferritin* [h1SMcG0007370] (FDR = 6.88151094599046e-13, log2FC = 1.23494736354217)
7. *Smed_AFG1-like ATPase* [h1SMcG0022995] (FDR = 0.00020406652654938, log2FC = 1.18699785154438)
8. *Smed_peptide methionine sulfoxide reductase* [h1SMcG0020033] (FDR = 3.60740799601908e-06, log2FC = 1.16641761504071)
9. *Smed_cytochrome P450 superfamily protein* [h1SMcG0007963] (FDR = 0.0387761213612101, log2FC = 1.06981857982849)
10. *Smed_DNA-directed RNA polymerase subunit RPB6* [h1SMcG0020367] (FDR = 0.0226477282486499, log2FC = 1.00357564613093)
11. *Smed_histone H1* [h1SMcG0008035] (FDR = 1.97822996494849e-08, log2FC = 0.980405995571909)
12. *Smed_trans-L-3-hydroxyproline dehydratase* [h1SMcG0015244] (FDR = 0.000345681243672477, log2FC = 0.964651374804372)
13. *Smed_S-adenosyl-L-methionine-dependent methyltransferase protein* [h1SMcG0000398] (FDR = 1.31026213522157e-07, log2FC = 0.963569899253442)
14. *Smed_S-adenosyl-L-methionine-dependent methyltransferase protein* [h1SMcG0000402] (FDR = 1.77985250747633e-09, log2FC = 0.930866944208397)
15. *Smed_40S ribosomal protein S27* [h1SMnG0025079] (FDR = 0.000105747602719927, log2FC = 0.928735946110406)
16. *Smed_40S ribosomal protein S21* [h1SMnG0034137] (FDR = 3.59678151846834e-08, log2FC = 0.895141925770015)
17. *Smed_PROG-1* [h1SMcG0005568] (FDR = 2.03525059543069e-10, log2FC = 0.888789130031133)
18. *Smed_ABC transporter C family protein* [h1SMcG0013272] (FDR = 0.0244442533123783, log2FC = 0.880436328840401)

19. *Smed_40S* ribosomal protein S26 [h1SMcG0004830] (FDR = 2.51602401233721e-05, log2FC = 0.880396140379722)
20. *Smed_delta-1* [h1SMcG0016126] (FDR = 0.00506634307162888, log2FC = 0.876636667231473)
21. *Smed_60S* ribosomal protein L37 [h1SMnG0009226] (FDR = 4.31252385355011e-06, log2FC = 0.874090290724696)
22. *Smed_60S* ribosomal protein L27a (uL15 family) [h1SMcG0001898] (FDR = 0.00506240720458914, log2FC = 0.868771837771833)
23. *Smed_Chibby* family protein [h1SMcG0014267] (FDR = 0.00420332432444453, log2FC = 0.860527105077816)
24. *Smed_Kunitz-type* serine protease inhibitor [h1SMnG0000851] (FDR = 0.00358892075963178, log2FC = 0.856747664769758)
25. *Smed_PROG-1* [h1SMcG0005614] (FDR = 4.45058020802247e-10, log2FC = 0.856201018783035)
26. *Smed_60S* acidic ribosomal protein P1 (P1/P2 family) [h1SMcG0014545] (FDR = 0.00801935340520298, log2FC = 0.853458787199681)
27. *Smed_ShK* domain-containing protein [h1SMnG0013594] (FDR = 0.00675970554979869, log2FC = 0.841950520033443)
28. *Smed_X1.D.A5.1* [h1SMnG0029262] (FDR = 0.0078287332265488, log2FC = 0.820196227209089)
29. *Smed_dynein* light chain [h1SMnG0019466] (FDR = 0.000984584956982265, log2FC = 0.820071939078189)
30. *Smed_40S* ribosomal protein S19 [h1SMcG0021578] (FDR = 8.14686614129657e-06, log2FC = 0.813967207633519)
31. *Smed_40S* ribosomal protein S2 [h1SMcG0011799] (FDR = 0.000908038517103189, log2FC = 0.810879312105277)
32. *Smed_40S* ribosomal protein S17 [h1SMcG0011238] (FDR = 0.00703929295231454, log2FC = 0.806454911155357)
33. *Smed_V-type* proton ATPase subunit 6 (V-ATPase G) [h1SMnG0017027] (FDR = 0.000666424622540026, log2FC = 0.801641419247056)
34. *Smed_Sec61γ* [h1SMnG0027394] (FDR = 0.0133339285383823, log2FC = 0.801494642483794)
35. *Smed_sterol* carrier protein 2-like [h1SMcG0002850] (FDR = 0.000264839650198437, log2FC = 0.801240476468087)
36. *Smed_EF-hand* calcium-binding domain profile protein [h1SMnG0006596] (FDR = 0.00273303525365074, log2FC = 0.785435852330556)
37. *Smed_Yip1* family protein [h1SMcG0015971] (FDR = 0.0109989205289308, log2FC = 0.765511065556372)
38. *Smed_slc16a-14* [h1SMcG0007904] (FDR = 0.000119227584305236, log2FC = 0.748676587419271)
39. *Smed_peroxisomal* membrane protein 4 [h1SMcG0020435] (FDR = 0.0462535414249423, log2FC = 0.748504856033728)
40. *Smed_deoxyuridine* triphosphatase [h1SMcG0004243] (FDR = 0.0200381588442557, log2FC = 0.740967113234474)
41. *Smed_60S* ribosomal protein L11-like (uL11 family) [h1SMcG0007885] (FDR = 0.000977908889009393, log2FC = 0.734688492746008)
42. *Smed_60S* ribosomal protein L30 [h1SMnG0034420] (FDR = 0.00140806386819594, log2FC = 0.734316146861171)

43. *Smed_lamin-like* [h1SMcG0007178] (FDR = 3.57091731687427e-05, log2FC = 0.724177539572768)
44. *Smed_40S ribosomal protein S15a* [h1SMcG0011437] (FDR = 0.00119550562634843, log2FC = 0.710883243952196)
45. *Smed_NB.21.11e* [h1SMcG0015271] (FDR = 0.000425907671016208, log2FC = 0.70928236564969)
46. *Smed_peptidyl-prolyl cis-trans isomerase-like* [h1SMcG0008138] (FDR = 3.24839034216188e-06, log2FC = 0.707547695976912)
47. *Smed_60S ribosomal protein L14* [h1SMcG0006469] (FDR = 0.0011556450197317, log2FC = 0.702712017735616)
48. *Smed_PROG-1* [h1SMcG0005567] (FDR = 2.44310759384651e-06, log2FC = 0.699986215526829)
49. *Smed_N-acetyltransferase 9-like protein* [h1SMcG0000224] (FDR = 0.0286293553670463, log2FC = 0.695029496997065)
50. *Smed_major facilitator superfamily domain-containing protein 6-like* [h1SMcG0014483] (FDR = 0.00231657614971256, log2FC = 0.693208936041522)
51. *Smed_B9 domain-containing protein 2-like* [h1SMcG0005869] (FDR = 0.0295467259823609, log2FC = 0.690818268472149)
52. *Smed_Plac8-C* [h1SMcG0005193] (FDR = 0.00174921382651123, log2FC = 0.686319641323166)
53. *Smed_caveolin-1* [h1SMcG0007790] (FDR = 0.0142393179677062, log2FC = 0.684129086118473)
54. *Smed_LITAF domain-containing protein* [h1SMcG0015131] (FDR = 0.0284838215975376, log2FC = 0.682129214536931)
55. *Smed_PROG-1* [h1SMcG0005613] (FDR = 2.93052145566199e-07, log2FC = 0.673537271408235)
56. *Smed_diamine acetyltransferase-like* [h1SMcG0002311] (FDR = 0.0224927937990014, log2FC = 0.66448965709154)
57. *Smed_VAMP family protein* [h1SMcG0019056] (FDR = 0.0108216370805176, log2FC = 0.660521388048831)
58. *Smed_PA28 (proteasome activator 11S regulator)* [h1SMcG0015959] (FDR = 0.0255016434802995, log2FC = 0.660160290011355)
59. *Smed_PROG-1* [h1SMcG0005540] (FDR = 7.10861698621884e-08, log2FC = 0.659336890946501)
60. *Smed_60S ribosomal protein L15* [h1SMcG0010975] (FDR = 2.30108543548415e-05, log2FC = 0.655545546352743)
61. *Smed_NME/NM23 nucleoside diphosphate kinase 1* [h1SMcG0006350] (FDR = 0.00509134972838611, log2FC = 0.652323989300503)
62. *Smed_60S ribosomal protein L27* [h1SMcG0002486] (FDR = 0.00392791659529735, log2FC = 0.650669857233817)
63. *Smed_mitochondrial pyruvate carrier 2-like* [h1SMnG0020099] (FDR = 0.033523696171513, log2FC = 0.640064497900294)
64. *Smed_FGFR1 oncogene partner/LisH domain-containing protein* [h1SMcG0008329] (FDR = 0.0255016434802995, log2FC = 0.634739392425108)
65. *Smed_ABCC/MRP-type ABC transporter* [h1SMcG0013268] (FDR = 0.00245915871545396, log2FC = 0.632448399107678)
66. *Smed_40S ribosomal protein S23* [h1SMcG0021056] (FDR = 0.0261744582914285, log2FC = 0.629800724503676)

67. *Smed_60S ribosomal protein L24* [h1SMcG0012146] (FDR = 8.09035314153778e-06, log2FC = 0.629344865413195)
68. *Smed_PEX11* [h1SMcG0014816] (FDR = 0.0187432726003942, log2FC = 0.628521511082218)
69. *Smed_40S ribosomal protein S16* [h1SMcG0006209] (FDR = 2.6536272391401e-05, log2FC = 0.628398487693793)
70. *Smed_40S ribosomal protein S24* [h1SMcG0021100] (FDR = 1.79619515338958e-07, log2FC = 0.62245065132686)
71. *Smed_mitochondrial ATP synthase subunit g* [h1SMnG0024674] (FDR = 0.00300790276083849, log2FC = 0.622102546912546)
72. *Smed_60S ribosomal protein L35a* [h1SMcG0001980] (FDR = 0.0156761838614002, log2FC = 0.6194568039793)
73. *Smed_sulfotransferase protein* [h1SMcG0020071] (FDR = 0.0031528415816333, log2FC = 0.617625518897966)
74. *Smed_HSP10* [h1SMnG0000109] (FDR = 0.0460475299033938, log2FC = 0.61631606570046)
75. *Smed_mpv17-like protein* [h1SMcG0012089] (FDR = 0.0145929993554335, log2FC = 0.615839511717404)
76. *Smed_slc6a-2* [h1SMcG0022500] (FDR = 0.0108216370805176, log2FC = 0.615495412006019)
77. *Smed_60S ribosomal protein L18* [h1SMcG0020762] (FDR = 0.00318468734019161, log2FC = 0.613798184079356)
78. *Smed_replication termination factor 2* [h1SMcG0012817] (FDR = 0.0019525753603805, log2FC = 0.613378904258802)
79. *Smed_40S ribosomal protein S6* [h1SMcG0005902] (FDR = 1.08757262233693e-05, log2FC = 0.611402759610424)
80. *Smed_ciliary microtubule inner protein 1-like* [h1SMcG0002797] (FDR = 0.0111597302272818, log2FC = 0.607940881250097)
81. *Smed_bolA-like* [h1SMcG0007128] (FDR = 0.00797372122261072, log2FC = 0.606624406924421)
82. *Smed_actin-like* [h1SMcG0014846] (FDR = 0.000674945499979479, log2FC = 0.606052504035403)
83. *Smed_40S ribosomal protein S14* [h1SMcG0013964] (FDR = 0.000258442040446408, log2FC = 0.604043532649951)
84. *Smed_carboxypeptidase-like protein* [h1SMcG0017024] (FDR = 0.0148091771668289, log2FC = 0.602290549226907)
85. *Smed_sulfotransferase protein* [h1SMcG0020315] (FDR = 0.000146835055979644, log2FC = 0.601457754589845)
86. *Smed_60S ribosomal protein L32* [h1SMcG0002815] (FDR = 0.000984584956982265, log2FC = 0.601379775929824)
87. *Smed_large ribosomal subunit protein eL28* [h1SMcG0019103] (FDR = 0.000137340537905367, log2FC = 0.599578740270803)
88. *Smed_ribosomal protein S7 (uS7/RPS5)* [h1SMcG0010916] (FDR = 0.00270689942468201, log2FC = 0.598226010332017)
89. *Smed_60S ribosomal protein L22* [h1SMcG0003147] (FDR = 0.00658496226242545, log2FC = 0.592931663340769)
90. *Smed_60S ribosomal protein L10a* [h1SMcG0018997] (FDR = 0.00242928289725995, log2FC = 0.592576226854946)

91. *Smed_NHP2-like protein (L7Ae family)* [h1SMcG0003059] (FDR = 0.0262657542010433, log2FC = 0.592203425206706)
92. *Smed_TNF receptor-associated factor-2-like protein A* [h1SMcG0018990] (FDR = 0.000573903600105424, log2FC = 0.592141874018859)
93. *Smed_GGCT family enzyme* [h1SMcG0022107] (FDR = 0.0108216370805176, log2FC = 0.589569419145976)
94. *Smed_actin-like* [h1SMcG0014849] (FDR = 0.000911645501866246, log2FC = 0.58737321364246)
95. *Smed_TCEN protein* [h1SMnG0012013] (FDR = 0.00275138792135578, log2FC = 0.58654300619284)

Downregulated genes at 36h (compared to earlier pre-hypoxia normoxia time-point of 0h) ranked from higher magnitude of downregulation:

1. *Smed_EGR-like protein 1* [h1SMcG0008212] (FDR = 5.29630944894902e-45, log2FC = -2.97912026846417)
2. *Smed_early growth response-3* [h1SMcG0008213] (FDR = 0.000208698923350804, log2FC = -2.80711495313152)
3. *Smed_ankyrin repeat domain-containing protein* [h1SMnG0032253] (FDR = 3.92754700514568e-05, log2FC = -2.52450461696032)
4. *Smed_Kunitz-type serine protease inhibitor* [h1SMcG0000534] (FDR = 6.34636576736954e-16, log2FC = -2.06302409243141)
5. *Smed_heat shock protein 70-2* [h1SMcG0009604] (FDR = 1.54972073683735e-39, log2FC = -2.05983220799762)
6. *Smed_heat shock protein 70-2* [h1SMcG0009603] (FDR = 3.60687625977299e-45, log2FC = -1.97871770821131)
7. *Smed_heat shock protein 70-2* [h1SMcG0009605] (FDR = 2.87146314460075e-46, log2FC = -1.97212184922629)
8. *Smed_small HSP protein* [h1SMcG0007183] (FDR = 5.10626252601581e-15, log2FC = -1.96431255513667)
9. *Smed_hadrian* [h1SMcG0020220] (FDR = 3.43901326613459e-07, log2FC = -1.91556333349366)
10. *Smed_diacylglycerol lipase* [h1SMnG0033599] (FDR = 0.00996240845642188, log2FC = -1.84181170388894)
11. *Smed_plasminogen-1* [h1SMcG0006227] (FDR = 0.000395907781303551, log2FC = -1.5092906267094)
12. *Smed_small heat shock protein (HSP20/alpha-crystallin family chaperone)* [h1SMcG0001680] (FDR = 0.0111597302272818, log2FC = -1.33791435748366)
13. *Smed_small HSP protein* [h1SMcG0001678] (FDR = 0.000220290389312962, log2FC = -1.28434246805508)
14. *Smed_TNF receptor-associated factor-2-like protein A* [h1SMcG0006232] (FDR = 8.49446260021418e-10, log2FC = -1.26850596027704)
15. *Smed_small HSP protein* [h1SMcG0001677] (FDR = 0.0467490088690481, log2FC = -1.26662919836495)
16. *Smed_small HSP protein* [h1SMcG0001675] (FDR = 0.0314947051011408, log2FC = -1.26451063083713)
17. *Smed_small HSP protein* [h1SMcG0001679] (FDR = 1.82379159213015e-05, log2FC = -1.25008699044861)

18. *Smed_gastric triacylglycerol lipase-like* [h1SMcG0006863] (FDR = 0.00794414038918635, log2FC = -1.23027189002777)
19. *Smed_slc7a-5* [h1SMcG0021830] (FDR = 0.000137401636992063, log2FC = -1.2290823431982)
20. *Smed_cytochrome P450 superfamily protein* [h1SMcG0010229] (FDR = 7.10861698621884e-08, log2FC = -1.19519260424263)
21. *Smed_small HSP protein* [h1SMcG0001676] (FDR= 8.44971181311284e-05, log2FC = -1.18972237142406)
22. *Smed_P450 2B-like protein* [h1SMcG0009082] (FDR = 0.0281180344725976, log2FC = -1.07165525606536)
23. *Smed_alkaline phosphatase-like* [h1SMcG0002552] (FDR = 0.0327826401594769, log2FC = -0.995094884139188)
24. *Smed_LTR retrotransposon Pol polyprotein* [h1SMcG0017867] (FDR = 0.0295467259823609, log2FC = -0.970200744554716)
25. *Smed_papain-like cysteine protease (cathepsin family)* [h1SMcG0017849] (FDR = 0.0318003912837573, log2FC = -0.936391055366393)
26. *Smed_slc18a-8* [h1SMcG0010030] (FDR = 0.0358729640596106, log2FC = -0.913476309109753)
27. *Smed_PH-like domain-containing protein* [h1SMcG0005736] (FDR = 3.19763779417833e-06, log2FC = -0.785495270653865)
28. *Smed_Hsp40* [h1SMcG0012561] (FDR = 0.000236206458276768, log2FC = -0.764084547766716)
29. *Smed_epidermal growth factor-1* [h1SMcG0022114] (FDR = 0.0467490088690481, log2FC = -0.739376386769367)
30. *Smed_strictosidine synthase-like* [h1SMcG0015357] (FDR = 0.00124129387824846, log2FC = -0.706971871606028)
31. *Smed_GMC family FAD-dependent oxidoreductase* [h1SMcG0009943] (FDR = 0.0382801228886335, log2FC = -0.701502665831085)
32. *Smed_IAP domain containing-1* [h1SMcG0007420] (FDR = 0.0118460837624835, log2FC = -0.687253858880749)
33. *Smed_slc4a-4* [h1SMcG0014894] (FDR = 0.000146835055979644, log2FC = -0.669900545855812)
34. *Smed_BAG domain-containing protein* [h1SMcG0004717] (FDR = 0.00844421293170983, log2FC = -0.666580007366716)
35. *Smed_peptidoglycan recognition protein-1* [h1SMcG0017863] (FDR = 0.0184716281213985, log2FC = -0.652117026868039)
36. *Smed_LCAT-related protein* [h1SMcG0020471] (FDR = 0.000265140435102602, log2FC = -0.646811166367524)
37. *Smed_ADAMTS domain-containing protein* [h1SMcG0018115] (FDR = 0.00216732164626776, log2FC = -0.643812292826791)
38. *Smed_WD40 repeat-containing protein* [h1SMcG0001457] (FDR = 3.98726679161835e-10, log2FC = -0.627706964249986)
39. *Smed_beta-1,3-N-acetylglucosaminyltransferase-like* [h1SMcG0022263] (FDR = 0.00329876541999725, log2FC = -0.62698621408773)
40. *Smed_sh3bp2* [h1SMcG0014255] (FDR = 0.0284838215975376, log2FC = -0.626084299414055)
41. *Smed_WD repeat and HMG-box DNA-binding protein* [h1SMcG0008938] (FDR = 0.00431705391366584, log2FC = -0.610689428662983)

42. *Smed_acetyl-coA synthetase (AMP-forming acyl-CoA synthetase family)* [h1SMcG0022274] (FDR = 5.39573997246625e-06, log2FC = -0.602799125114474)
43. *Smed_90 kDa heat shock protein* [h1SMcG0005587] (FDR = 1.45613612777136e-05, log2FC = -0.590570653871901)
44. *Smed_class B secretin-like GPCR* [h1SMcG0005956] (FDR = 5.84474922081613e-05, log2FC = -0.590489544169888)

3h vs 12h (FDR = 0.05, FC = 1.5)

Upregulated genes at 12h (compared to earlier hypoxic time-point of 3h), excluding 'ubiquitin' and 'polyubiquitin' hits (h1SMcG0020968, h1SMcG0020970, h1SMcG0020975, h1SMcG0020969, h1SMcG0020974, h1SMcG0020967), ranked from higher magnitude of upregulation:

1. *Smed_TNF receptor associated factor 2* [h1SMcG0022817] (FDR = 2.74278373001003e-35, log2FC = 5.04394378537694)
2. *Smed_cellular IAP repeat-containing protein* [h1SMcG0001546] (FDR = 2.17725612290239e-16, log2FC = 2.81635963242565)
3. *Smed_Plac8-E* [h1SMcG0005183] (FDR = 9.25001567439503e-07, log2FC = 2.07782305982806)
4. *Smed_klf10/11* [h1SMcG0018872] (FDR = 7.2403833404321e-45, log2FC = 1.76620847476347)
5. *Smed_cytochrome P450 2K1-like protein* [h1SMcG0010360] (FDR = 0.00211225519077863, log2FC = 1.63356541433809)
6. *Smed_histone H3-like* [h1SMcG0007872] (FDR = 0.00072809834319238, log2FC = 1.57005236416942)
7. *Smed_protein phosphatase 1 regulatory-1* [h1SMcG0010862] (FDR = 5.2880730759311e-16, log2FC = 1.46783101597339)
8. *Smed_IAP domain-containing protein* [h1SMcG0007403] (FDR = 4.46246942558708e-13, log2FC = 1.39086611915001)
9. *Smed_histone H1* [h1SMcG0008035] (FDR = 7.69979012722946e-37, log2FC = 1.15722352500149)
10. *Smed_histone 2A-like* [h1SMcG0014269] (FDR = 0.0424805325353168, log2FC = 1.15320672699798)
11. *Smed_TNF receptor-associated factor* [h1SMcG0007071] (FDR = 2.15057857196629e-16, log2FC = 1.14971399227507)
12. *Smed_E3 ubiquitin-protein ligase ARIH1* [h1SMcG0021303] (FDR = 1.42393217894396e-36, log2FC = 1.14749468369257)
13. *Smed_slc6a-24* [h1SMnG0009932] (FDR = 0.0121850695771528, log2FC = 1.1060376728488)
14. *Smed_PIM-1* [h1SMcG0003234] (FDR = 1.16324820477513e-10, log2FC = 0.98023983255778)
15. *Smed_ectopic P granules protein 5 homolog* [h1SMcG0003834] (FDR = 3.6593235941213e-05, log2FC = 0.95671192889445)

16. *Smed_soma* ferritin [h1SMcG0007370] (FDR = 7.2131437733521e-13, log2FC = 0.94200135086997)
17. *Smed_peptide* methionine sulfoxide reductase [h1SMcG0020033] (FDR = 5.80835687638959e-06, log2FC = 0.921786457449023)
18. *Smed_folliculin-interacting* protein 2-like [h1SMcG0006386] (FDR = 9.63142002719171e-10, log2FC = 0.860393104378421)
19. *Smed_slc20a-2* [h1SMcG0001112] (FDR = 0.00241987093910656, log2FC = 0.822268722423353)
20. *Smed_TNF* receptor-associated factor 2 [h1SMcG0015086] (FDR = 0.0259919567084051, log2FC = 0.817507289185992)
21. *Smed_ferritin* [h1SMcG0007372] (FDR = 6.3471827740735e-12, log2FC = 0.793534715200956)
22. *Smed_protein* phosphatase 1 regulatory subunit [h1SMcG0019740] (FDR = 4.62769271610731e-10, log2FC = 0.769057331577846)
23. *Smed_E3* ubiquitin ligase RBR family protein [h1SMcG0004408] (FDR = 1.40840079680821e-15, log2FC = 0.757135427209915)
24. *Smed_D-3-phosphoglycerate* dehydrogenase-like [h1SMcG0017641] (FDR = 1.28214184065436e-09, log2FC = 0.740431371440956)
25. *Smed_RING-type* E3 ubiquitin-protein ligase [h1SMcG0018197] (FDR = 0.0254369255326289, log2FC = 0.672143077840019)
26. *Smed_pantothenate* kinase protein [h1SMcG0016781] (FDR = 1.16324820477513e-10, log2FC = 0.641563989417402)
27. *Smed_protein* phosphatase 1 regulatory subunit 3-like [h1SMcG0004471] (FDR = 3.73237439458067e-10, log2FC = 0.633124144380138)
28. *Smed_trans-L-3-hydroxyproline* dehydratase [h1SMcG0015244] (FDR = 0.0120817637890756, log2FC = 0.591624454821761)
29. *Smed_growth* factor receptor-bound protein [h1SMcG0006330] (FDR = 0.0122910285262877, log2FC = 0.590587578125572)
30. *Smed_RMI1-like* [h1SMcG0015397] (FDR = 0.0287882418979312, log2FC = 0.585469524020475)

Downregulated genes at 12h (compared to earlier hypoxic time-point of 3h), ranked from higher magnitude of downregulation:

1. *Smed_early growth response-3* [h1SMcG0008213] (FDR = 3.04900140425764e-08, log2FC = -1.37204888552024)
2. *Smed_small* subunit processome component 20 homolog [h1SMcG0004349] (FDR = 7.32128768398014e-09, log2FC = -1.35523816158727)
3. *Smed_EGR-like protein 1* [h1SMcG0008212] (FDR = 9.65659935634157e-20, log2FC = -1.2546614552316)
4. *Smed_slc18a-8* [h1SMcG0010030] (FDR = 0.0220134250499122, log2FC = -1.06849580712239)
5. *Smed_TNF* receptor associated factor-2 [h1SMcG0019430] (FDR = 7.0057579164187e-06, log2FC = -0.91340130931029)
6. *Smed_transposon* Ty3-G Gag-Pol polyprotein [h1SMcG0017868] (FDR = 0.0452389869842475, log2FC = -0.798624165802195)

7. *Smed_PH-like domain-containing protein* [h1SMcG0005736] (FDR = 1.61323479485798e-11, log2FC = -0.766059169463975)
8. *Smed_90 kDa heat shock protein* [h1SMcG0005587] (FDR = 6.2586636167989e-08, log2FC = -0.701427226351358)
9. *Smed_SEZ6/CSMD family protein* [h1SMcG0003529] (FDR = 0.0134783002193715, log2FC = -0.646652185582706)
10. *Smed_hadrian* [h1SMcG0020220] (FDR = 0.0224278884754696, log2FC = -0.63335432734956)

3h vs 24h (FDR = 0.05, FC = 1.5)

Upregulated genes at 24h (compared to earlier hypoxic time-point of 3h), excluding 'ubiquitin' and 'polyubiquitin' hits (h1SMcG0020968, h1SMcG0020974, h1SMcG0020969, h1SMcG0020966, h1SMcG0020967, h1SMcG0020975, h1SMcG0020970, h1SMcG0020973), ranked from higher magnitude of upregulation:

1. *Smed_cellular IAP repeat-containing protein* [h1SMcG0001546] (FDR = 5.38831833931124e-25, log2FC = 4.48980594553004)
2. *Smed_soma ferritin* [h1SMcG0007370] (FDR = 1.09771580892436e-81, log2FC = 2.38170045425925)
3. *Smed_E3 ubiquitin-protein ligase ARIH1* [h1SMcG0021303] (FDR = 5.84180049108717e-121, log2FC = 2.28310250196295)
4. *Smed_TNF receptor-associated factor 2* [h1SMcG0015086] (FDR = 3.42929998320591e-33, log2FC = 2.27495224560965)
5. *Smed_E3 ubiquitin ligase RBR family protein* [h1SMcG0004408] (FDR = 8.53656586600655e-187, log2FC = 2.04155897005111)
6. *Smed_ferritin* [h1SMcG0007372] (FDR = 9.55170279797506e-51, log2FC = 1.99939707495997)
7. *Smed_pantothenate kinase protein* [h1SMcG0016781] (FDR = 3.88127174454278e-49, log2FC = 1.82919766161524)
8. *Smed_peptide methionine sulfoxide reductase* [h1SMcG0020033] (FDR = 5.2058172763782e-29, log2FC = 1.74054509105932)
9. *Smed_klf10/11* [h1SMcG0018872] (FDR = 9.34777701054241e-39, log2FC = 1.64618551706159)
10. *Smed_IAP domain-containing protein* [h1SMcG0007403] (FDR = 0.000120332556404965, log2FC = 1.6267518724941)
11. *Smed_TNF receptor-associated factor* [h1SMcG0007071] (FDR = 6.96232479310833e-37, log2FC = 1.6110591930566)
12. *Smed_histone H3-like* [h1SMcG0007872] (FDR = 0.00700828870119228, log2FC = 1.5007030355138)
13. *Smed_Plac8-E* [h1SMcG0005183] (FDR = 0.0299564445864501, log2FC = 1.37449756845093)
14. *Smed_pantothenate kinase protein* [h1SMcG0016782] (FDR = 1.25097557384644e-27, log2FC = 1.35014958007983)

15. **Smed_NSD-1** [h1SMcG0019813] (FDR = 0.0229963165482977, log2FC = 1.31712480399292)
16. **Smed_slc20a-2** [h1SMcG0001112] (FDR = 1.31007703860405e-08, log2FC = 1.29667263824922)
17. *Smed_P2X* purinergic receptor [h1SMnG0009965] (FDR = 0.029125844926004, log2FC = 1.26233326760501)
18. *Smed_cytochrome P450 2K1-like protein* [h1SMcG0010360] (FDR = 1.69891342599146e-19, log2FC = 1.22610189424833)
19. *Smed_60S* ribosomal protein L30 [h1SMcG0003843] (FDR = 0.00415666313834432, log2FC = 1.12244863805394)
20. **Smed_histone H1** [h1SMcG0008035] (FDR= 6.32002333917716e-18, log2FC= 1.1140710740225)
21. **Smed_epidermal growth factor-1** [h1SMcG0022114] (FDR = 1.27551823217999e-05, log2FC = 1.09174118055106)
22. **Smed_slc24a-5** [h1SMnG0034353] (FDR = 1.51145311692605e-06, log2FC = 1.08681360816988)
23. *Smed_sulfotransferase* protein [h1SMcG0008457] (FDR = 0.0002395303128148, log2FC = 1.06771978215229)
24. *Smed_protein phosphatase 1 regulatory subunit* [h1SMcG0019740] (FDR = 8.80560386200506e-18, log2FC = 0.992757761662251)
25. *Smed_major facilitator superfamily domain-containing protein 6-like* [h1SMcG0014483] (FDR = 4.71194105132941e-12, log2FC = 0.969903966341252)
26. *Smed_universal stress protein* [h1SMcG0009643] (FDR = 3.97566082535641e-15, log2FC = 0.968988482278447)
27. *Smed_integrase core domain-containing protein* [h1SMnG0030698] (FDR = 0.0224990964814773, log2FC = 0.966321880567578)
28. **Smed_PIM-1** [h1SMcG0003234] (FDR = 0.00868268423728287, log2FC = 0.948610427972408)
29. *Smed_monocarboxylate transporter protein* [h1SMcG0016325] (FDR = 0.00123744962647504, log2FC = 0.909484520397574)
30. *Smed_apolipophorins-like* [h1SMcG0002269] (FDR = 0.00860900818381748, log2FC = 0.871932543283345)
31. *Smed_trans-L-3-hydroxyproline dehydratase* [h1SMcG0015244] (FDR = 0.000219199935988064, log2FC = 0.859077599317299)
32. *Smed_receptor-transporting protein* [h1SMcG0006507] (FDR = 1.27551823217999e-05, log2FC = 0.854104528560681)
33. *Smed_basic-leucine zipper (bZIP) domain transcription factor* [h1SMnG0004780] (FDR = 0.000753193739963317, log2FC = 0.83319341483128)
34. *Smed_small HSP protein* [h1SMcG0001680] (FDR= 1.27551823217999e-05, log2FC = 0.817060107649416)
35. *Smed_protein phosphatase 1 regulatory subunit 3-like* [h1SMcG0004471] (FDR = 2.61910300919782e-13, log2FC = 0.812649344667087)
36. *Smed_mpv17-like protein* [h1SMcG0021924] (FDR = 3.51559369201611e-05, log2FC = 0.803775392128456)
37. *Smed_small HSP protein* [h1SMcG0001675] (FDR = 2.57232767080393e-06, log2FC = 0.802336806714596)
38. *Smed_small HSP protein* [h1SMcG0001678] (FDR= 0.0150405290103757, log2FC= 0.802296757783083)

39. *Smed_adiponectin receptor protein-like* [h1SMcG0013451] (FDR = 1.85579147316627e-20, log2FC = 0.789439269479018)
40. *Smed_5'-AMP-activated protein kinase, gamma regulatory subunit* [h1SMcG0005059] (FDR = 1.29229337736078e-15, log2FC = 0.788804920521244)
41. *Smed_ferritin-like protein* [h1SMcG0011235] (FDR = 1.50251922839502e-06, log2FC = 0.760535697524752)
42. *Smed_GGCT family enzyme* [h1SMcG0022107] (FDR = 0.0389995878529196, log2FC = 0.745882999521717)
43. *Smed_small HSP protein* [h1SMcG0001677] (FDR = 0.00227381925480665, log2FC = 0.743831243775877)
44. *Smed_plakin family cytoskeletal linker protein* [h1SMcG0000479] (FDR = 0.00784388483137106, log2FC = 0.73161306913456)
45. *Smed_LRR-containing protein* [h1SMcG0015004] (FDR = 4.73200565846625e-09, log2FC = 0.727658910643335)
46. *Smed_growth factor receptor-bound protein* [h1SMcG0006330] (FDR = 0.00443206074265776, log2FC = 0.727240493545789)
47. *Smed_cramped-like* [h1SMcG0002801] (FDR = 0.0489606996511945, log2FC = 0.727217140533045)
48. *Smed_MAPK phosphatase* [h1SMcG0019962] (FDR = 4.07934694078716e-10, log2FC = 0.721971664733777)
49. *Smed_slc25a-29* [h1SMcG0020425] (FDR = 2.16886754362083e-07, log2FC = 0.702649924248855)
50. *Smed_cytochrome P450 2B-like protein* [>SMEST072744002.1.p1 type:complete len:507 gc:universal SMEST072744002.1:392-1912(+)] (FDR = 1.32401878257203e-06, log2FC = 0.702190252488643)
51. *Smed_protein phosphatase 1 regulatory-1* [h1SMcG0010862] (FDR = 0.010319392294908, log2FC = 0.692766753743603)
52. *Smed_ABCD4-like ABC transporter* [h1SMcG0002437] (FDR = 2.46853630128244e-05, log2FC = 0.674954111815164)
53. *Smed_ABCC/MRP-type ATP-binding cassette transporter* [h1SMcG0013279] (FDR = 0.00180361275204109, log2FC = 0.656880514460448)
54. *Smed_atf2-1* [h1SMcG0001134] (FDR = 3.50118002407489e-05, log2FC = 0.641731660785833)
55. *Smed_C2H2-type zinc finger domain-containing protein* [h1SMcG0000013] (FDR = 0.00542486263746407, log2FC = 0.641184181051634)
56. *Smed_phospholipase A2-like* [h1SMcG0017624] (FDR = 0.000886740862810799, log2FC = 0.629034665038605)
57. *Smed_D-3-phosphoglycerate dehydrogenase-like* [h1SMcG0017641] (FDR = 0.000303485468588061, log2FC = 0.624310309286773)
58. *Smed_FAD-dependent oxidoreductase domain-containing protein 1-like* [h1SMcG0004813] (FDR = 1.55228350512167e-05, log2FC = 0.623921413996491)
59. *Smed_M20 family zinc-dependent metalloproteinase* [h1SMnG0005218] (FDR = 0.00746117910412516, log2FC = 0.62349582374789)
60. *Smed_P450 2B-like protein* [h1SMcG0009079] (FDR = 0.000463839615397642, log2FC = 0.616876399713567)
61. *Smed_ankyrin repeat and zinc finger domain-containing protein* [h1SMcG0022488] (FDR = 7.97280579672435e-09, log2FC = 0.613476026967275)

62. **Smed_1-Jun** [h1SMcG0021300] (FDR = 4.80090539584992e-06, log2FC = 0.610276558947562)
63. *Smed_ABCC* (MRP-type) ATP-binding cassette transporter [h1SMcG0020539] (FDR = 0.0226807348927802, log2FC = 0.607902355868411)
64. *Smed_ABHD* superfamily protein [h1SMcG0011435] (FDR = 6.84224317944876e-05, log2FC = 0.600618824884894)
65. **Smed_slc6a-2** [h1SMcG0022500] (FDR = 0.00986896983332016, log2FC = 0.599853127884372)
66. *Smed_NAD(P)*-dependent oxidoreductase [h1SMcG0007023] (FDR = 0.00200574770331279, log2FC = 0.596074757264264)

Downregulated genes at 24h (compared to earlier hypoxic time-point of 3h), ranked from higher magnitude of downregulation:

1. *Smed_TBC1* domain-containing Rab GTPase-activating protein [h1SMcG0023066] (FDR = 0.0123608533779445, log2FC = -2.49384188556918)
2. **Smed_early growth response-3** [h1SMcG0008213] (FDR = 6.24544641693867e-06, log2FC = -1.75964240757588)
3. **Smed_EGR-like protein 1** [h1SMcG0008212] (FDR = 3.41521712725604e-13, log2FC = -1.5089344239203)
4. *Smed_PH*-like domain-containing protein [h1SMcG0005736] (FDR = 6.560415103527e-12, log2FC = -1.21484147272132)
5. **Smed_hadrian** [h1SMcG0020220] (FDR = 0.00700828870119228, log2FC = -1.17881579728274)
6. **Smed_tissue factor pathway inhibitor-1** [h1SMnG0000861] (FDR = 0.00887193630085529, log2FC = -1.13858578507958)
7. *Smed_cytochrome P450* superfamily protein [h1SMcG0010229] (FDR = 4.33017244323945e-05, log2FC = -1.03866869422161)
8. *Smed_sulfotransferase* protein [h1SMcG0020062] (FDR = 0.00230631861644892, log2FC = -0.972140296981768)
9. *Smed_Kunitz*-type serine protease inhibitor [h1SMcG0000534] (FDR = 0.0147869557421846, log2FC = -0.969197688988542)
10. **Smed_slc7a-5** [h1SMcG0021830] (FDR = 0.00443206074265776, log2FC = -0.942239055145818)
11. *Smed_short-chain dehydrogenase/reductase (SDR)* family protein [h1SMcG0010682] (FDR = 0.0149239993707044, log2FC = -0.918733944336265)
12. *Smed_beta-1,3-N-acetylglucosaminyltransferase*-like [h1SMcG0022263] (FDR = 2.47921335174805e-06, log2FC = -0.812683605590511)
13. *Smed_M13* family neprilysin-like zinc metalloprotease [h1SMcG0000523] (FDR = 0.0235201943569632, log2FC = -0.794683507884236)
14. *Smed_alkaline phosphatase*-like [h1SMcG0002566] (FDR = 0.00553031927024381, log2FC = -0.721829679342273)
15. *Smed_ADAMTS* domain-containing protein [h1SMcG0018115] (FDR = 6.8798840900601e-07, log2FC = -0.683965998838979)
16. *Smed_GMC* family FAD-dependent oxidoreductase [h1SMcG0009943] (FDR = 0.026916246351096, log2FC = -0.665073777917793)

17. *Smed_SAMHD1-like* [h1SMcG0009332] (FDR = 0.000886740862810799, log2FC = -0.601142125824112)
18. *Smed_sh3bp2* [h1SMcG0014255] (FDR = 0.00262691052068664, log2FC = -0.594261968212882)
19. *Smed_diacylglycerol kinase protein* [h1SMcG0000617] (FDR = 0.0167152427553759, log2FC = -0.593313540208783)

3h vs 36h (FDR = 0.05, FC = 1.5)

Upregulated genes at 36h (compared to earlier hypoxic time-point of 3h), excluding 'ubiquitin' and 'polyubiquitin' hits (h1SMcG0020968, h1SMcG0020975, h1SMcG0020970, h1SMcG0020967, h1SMcG0020974, and h1SMcG0020969), ranked from higher magnitude of upregulation:

1. *Smed_sulfotransferase protein* [h1SMcG0008457] (FDR = 3.8830765922432e-20, log2FC = 1.85389304769923)
2. *Smed_small HSP protein* [h1SMcG0001680] (FDR= 6.30424908136116e-69, log2FC = 1.48476448585517)
3. *Smed_small HSP protein* [h1SMcG0001675] (FDR = 1.59825601964084e-55, log2FC = 1.42035423951605)
4. *Smed_small HSP protein* [h1SMcG0001677] (FDR= 1.92662339122105e-27, log2FC= 1.41061539047926)
5. *Smed_small HSP protein* [h1SMcG0001676] (FDR= 9.90538668670149e-25, log2FC= 1.33226650642306)
6. *Smed_small HSP protein* [h1SMcG0001679] (FDR= 4.22782364803743e-13, log2FC= 1.25342148348441)
7. *Smed_small HSP protein* [h1SMcG0001678] (FDR= 3.09595950403573e-10, log2FC= 1.12561414493191)
8. *Smed_ABC transporter C family protein* [h1SMcG0013272] (FDR = 0.00161085175717368, log2FC = 0.995495812036281)
9. *Smed_ferritin* [h1SMcG0007372] (FDR = 3.24982600580225e-37, log2FC = 0.953346035597735)
10. *Smed_soma ferritin* [h1SMcG0007370] (FDR = 7.63224973514201e-19, log2FC = 0.939564543115073)
11. *Smed_ABCC/MRP-type ABC transporter* [h1SMcG0013268] (FDR = 5.03880690047621e-15, log2FC = 0.925348216532546)
12. *Smed_S-adenosyl-L-methionine-dependent methyltransferase protein* [h1SMcG0000402] (FDR = 1.09506075177893e-15, log2FC = 0.82872936270782)
13. *Smed_slc16a-14* [h1SMcG0007904] (FDR = 2.24788817455771e-11, log2FC = 0.827108710849844)
14. *Smed_delta-1* [h1SMcG0016126] (FDR = 0.0332207228929974, log2FC = 0.808826637156087)
15. *Smed_actin-like protein* [h1SMcG0014849] (FDR = 2.77087602960791e-10, log2FC = 0.790077305832681)
16. *Smed_slc6a-2* [h1SMcG0022500] (FDR = 9.59609367129782e-08, log2FC = 0.739888467655888)

17. *Smed_ABCC/MRP-type ABC transporter* [h1SMcG0013273] (FDR = 0.00121568692332062, log2FC = 0.719293157167935)
18. *Smed_peptide methionine sulfoxide reductase* [h1SMcG0020033] (FDR = 0.00506422118221867, log2FC = 0.71230655172446)
19. *Smed_ABCC/MRP-type ATP-binding cassette transporter* [h1SMcG0013279] (FDR = 1.05169432383295e-06, log2FC = 0.699847768218244)
20. *Smed_atf2-1* [h1SMcG0001134] (FDR = 1.26125023645885e-09, log2FC = 0.695435839926572)
21. *Smed_major facilitator superfamily domain-containing protein 6-like* [h1SMcG0014483] (FDR = 3.13689283793716e-07, log2FC = 0.684193471668753)
22. *Smed_GGCT family enzyme* [h1SMcG0022107] (FDR = 0.000320174055510822, log2FC = 0.681183044250753)
23. *Smed_diamine acetyltransferase-like* [h1SMcG0002311] (FDR = 1.7056272348204e-05, log2FC = 0.665157998918815)
24. *Smed_S-adenosyl-L-methionine-dependent methyltransferase protein* [h1SMcG0000398] (FDR = 4.08624127553478e-10, log2FC = 0.658806813502815)
25. *Smed_PROG-1* [h1SMcG0005614] (FDR = 6.35333684648321e-10, log2FC = 0.643075896183839)
26. *Smed_lamin-like* [h1SMcG0007178] (FDR = 2.31582688383738e-16, log2FC = 0.637364698487646)
27. *Smed_Mp-2* [h1SMcG0011793] (FDR = 0.0233231868966012, log2FC = 0.608340740522475)
28. *Smed_90 kDa heat shock protein* [h1SMcG0005587] (FDR = 1.00826229137927e-06, log2FC = 0.608152014166547)

Downregulated genes at 36h (compared to earlier hypoxic time-point of 3h), ranked from higher magnitude of downregulation:

1. *Smed_Kunitz-type serine protease inhibitor* [h1SMcG0000534] (FDR = 2.58527058157233e-15, log2FC = -1.99176136120191)
2. *Smed_cytochrome P450 2K1-like protein* [h1SMcG0010360] (FDR = 0.0430351921913951, log2FC = -1.9439996160378)
3. *Smed_hadrian* [h1SMcG0020220] (FDR = 1.30343246825908e-14, log2FC = -1.87670269967454)
4. *Smed_EGR-like protein 1* [h1SMcG0008212] (FDR = 4.08766172279297e-30, log2FC = -1.86594140606607)
5. *Smed_early growth response-3* [h1SMcG0008213] (FDR = 3.09770049280271e-06, log2FC = -1.84140961544182)
6. *Smed_tissue factor pathway inhibitor-1* [h1SMnG0000861] (FDR = 7.84558603218898e-05, log2FC = -1.48613675876164)
7. *Smed_TNF receptor-associated factor-2-like protein A* [h1SMcG0006232] (FDR = 6.58310920777223e-25, log2FC = -1.36285501735358)
8. *Smed_klf10/11* [h1SMcG0018872] (FDR = 3.87289783717584e-26, log2FC = -1.34324380880539)
9. *Smed_tubulin alpha chain* [h1SMcG0017866] (FDR = 0.000444228840385723, log2FC = -1.29892828655969)
10. *Smed_plasminogen-1* [h1SMcG0006227] (FDR = 0.0062940865915385, log2FC = -1.27395930995057)

11. **Smed_slc7a-5** [h1SMcG0021830] (FDR = 6.16810115596807e-05, log2FC = -1.21980469042418)
12. *Smed_alkaline phosphatase-like* [h1SMcG0002552] (FDR = 4.94922607924005e-09, log2FC = -1.15696640547563)
13. *Smed_cytochrome P450 superfamily protein* [h1SMcG0010229] (FDR = 1.50148110465329e-07, log2FC = -1.15284315652207)
14. *Smed_gastric triacylglycerol lipase-like* [h1SMcG0006863] (FDR = 0.000972521070998486, log2FC = -1.13743385195866)
15. **Smed_blitzschnell 3G** [h1SMcG0017123] (FDR = 0.000120847876942397, log2FC = -0.992710742591892)
16. *Smed_P450 2B-like protein* [h1SMcG0009082] (FDR = 0.0400114983854911, log2FC = -0.919455167861901)
17. **Smed_slc30a-2** [h1SMcG0000860] (FDR = 0.00211493622381286, log2FC = -0.893666526082432)
18. **Smed_epidermal growth factor-1** [h1SMcG0022114] (FDR = 0.0164854182523812, log2FC = -0.874450366671123)
19. *Smed_GNAT family N-acetyltransferase* [h1SMcG0018715] (FDR = 0.000193978723383064, log2FC = -0.835112870240822)
20. *Smed_LCAT-related protein* [h1SMcG0020471] (FDR = 1.13386000887158e-10, log2FC = -0.809802774733136)
21. **Smed_peptidoglycan recognition protein-1** [h1SMcG0017862] (FDR = 0.000122120591415258, log2FC = -0.746096684884699)
22. *Smed_PH-like domain-containing protein* [h1SMcG0005736] (FDR = 1.2070196977303e-11, log2FC = -0.740252159923489)
23. *Smed_Ataxin-1-like* [h1SMcG0020563] (FDR = 0.000325346292971029, log2FC = -0.735783240763506)
24. *Smed_VWFC domain-containing protein* [h1SMcG0004289] (FDR = 0.00692752943049758, log2FC = -0.726394423272275)
25. *Smed_GMC family FAD-dependent oxidoreductase* [h1SMcG0009943] (FDR = 0.0158955640897263, log2FC = -0.715455415202847)
26. *Smed_alkaline phosphatase-like* [h1SMcG0002566] (FDR = 0.0319927436316347, log2FC = -0.715108514234498)
27. **Smed_IAP domain containing-1** [h1SMcG0007420] (FDR = 0.000320223540523946, log2FC = -0.690105789048728)
28. *Smed_metallophosphoesterase (ADP-ribose/CDP-alcohol diphosphatase-like)* [h1SMcG0007008] (FDR = 1.39309366571455e-05, log2FC = -0.684000948240669)
29. *Smed_TNF receptor-associated protein* [h1SMcG0014828] (FDR = 9.11195488001358e-05, log2FC = -0.682137278312866)
30. **Smed_protein phosphatase 1 regulatory-1** [h1SMcG0010862] (FDR = 0.00480184865038843, log2FC = -0.676109007195549)
31. **Smed_serine/threonine protein kinase** [h1SMcG0007377] (FDR = 0.00400246147459131, log2FC = -0.66418082714729)
32. *Smed_ankyrin repeat-containing protein* [h1SMnG0032058] (FDR = 0.0416966689594186, log2FC = -0.651078489691718)
33. *Smed_strictosidine synthase-like* [h1SMcG0015357] (FDR = 0.00047093841497938, log2FC = -0.650004881529721)
34. **Smed_sh3bp2** [h1SMcG0014255] (FDR = 0.00146640253752241, log2FC = -0.640246244920911)

35. **Smed_TNF receptor associated factor-2** [h1SMcG0019430] (FDR = 0.0279529891426486, log2FC = -0.629109633771613)
36. *Smed_serine/threonine-protein kinase* [h1SMcG0017420] (FDR = 0.00050008312397935, log2FC = -0.619551923704039)
37. *Smed_family A GPCR* [h1SMcG0018594] (FDR = 2.83092500499683e-06, log2FC = -0.60551571502532)
38. *Smed_GNAT family N-acetyltransferase* [h1SMcG0018321] (FDR = 3.8511575712805e-06, log2FC = -0.600324734047331)
39. *Smed_TtcA-like* [h1SMcG0000801] (FDR = 0.0323388647888916, log2FC = -0.592031909163593)

12h vs 24h (FC = 1.5, FDR 0.05)

Upregulated genes at 24h (compared to earlier time-point of 12h) ranked from higher magnitude of upregulation, excluding 'ubiquitin' and 'polyubiquitin' hits (h1SMcG0020974, h1SMcG0020967, h1SMcG0020966, h1SMcG0020969, h1SMcG0020968, h1SMcG0020975, h1SMcG0020973, h1SMcG0020970 and h1SMcG0020971):

1. *Smed_cellular IAP repeat-containing protein* [h1SMcG0001546] (FDR = 0.00115802123632897, log2FC = 1.67174943157185)
2. **Smed_TNF receptor-associated factor 2** [h1SMcG0015086] (FDR = 3.04826482967968e-15, log2FC = 1.45578393784975)
3. *Smed_soma ferritin* [h1SMcG0007370] (FDR = 1.29694626242424e-22, log2FC = 1.4380976756842)
4. *Smed_E3 ubiquitin ligase RBR family protein* [h1SMcG0004408] (FDR = 9.49499108181292e-35, log2FC = 1.28296646534147)
5. *Smed_ferritin* [h1SMcG0007372] (FDR = 7.41221311607049e-15, log2FC = 1.20409441930868)
6. *Smed_pantothenate kinase protein* [h1SMcG0016781] (FDR = 2.40005711327688e-21, log2FC = 1.18580500342998)
7. *Smed_small subunit processome component 20 homolog* [h1SMcG0004349] (FDR = 0.0234436824053391, log2FC = 1.17152208654587)
8. *Smed_ankyrin repeat domain-containing protein* [h1SMcG0029278] (FDR = 6.0280156171846e-05, log2FC = 1.15668944899923)
9. *Smed_E3 ubiquitin-protein ligase ARIH1* [h1SMcG0021303] (FDR = 2.02255410363611e-29, log2FC = 1.13413550637267)
10. *Smed_sulfotransferase protein* [h1SMcG0008457] (FDR = 0.00179245897996835, log2FC = 1.03220066715605)
11. *Smed_pantothenate kinase protein* [h1SMcG0016782] (FDR = 1.20796660649376e-11, log2FC = 0.832205322392106)
12. *Smed_peptide methionine sulfoxide reductase* [h1SMcG0020033] (FDR = 3.29449793789198e-07, log2FC = 0.817120064732143)
13. *Smed_C2H2-type zinc finger domain-containing protein* [h1SMcG0000013] (FDR = 0.00435144853613636, log2FC = 0.719088550447476)
14. *Smed_ABCC/MRP-type ATP-binding cassette transporter* [h1SMcG0013279] (FDR = 0.000883395585949932, log2FC = 0.666196353788647)

15. **Smed_epidermal growth factor-1** [h1SMcG0022114] (FDR = 0.0048936459923568, log2FC = 0.624641711702953)
16. **Smed_SOSd** [h1SMcG0011383] (FDR = 0.032487025969372, log2FC = 0.607637724949078)

Downregulated genes at 24h (compared to earlier time-point of 12h) ranked from higher magnitude of downregulation:

1. *Smed_P-type ATPase* [h1SMcG0023126] (FDR = 8.29901047929455e-11, log2FC = -5.78777080088881)
2. *Smed_Rho GTPase-activating protein (RhoGAP) family* [h1SMcG0010753] (FDR = 0.00920415478115433, log2FC = -1.68693833104466)
3. *Smed_peptidase M8 family metalloprotease* [h1SMcG0008332] (FDR = 0.0458246644037696, log2FC = -0.989911296078592)
4. **Smed_protein phosphatase 1 regulatory-1** [h1SMcG0010862] (FDR = 0.00251748915732401, log2FC = -0.777012183138519)
5. *Smed_alpha-tocopherol transfer protein* [h1SMcG0022485] (FDR = 0.0078520793405671, log2FC = -0.735518501309427)
6. *Smed_beta-1,3-N-acetylglucosaminyltransferase-like* [h1SMcG0022263] (FDR = 0.0203796283972447, log2FC = -0.599456424342842)

12h vs 36h (FDR = 0.05, FC = 1.5)

Upregulated genes at 36h (compared to earlier hypoxic time-point of 12h), excluding 'ubiquitin' and 'polyubiquitin' hits (including h1SMcG0017816) ranked from higher magnitude of upregulation:

1. *Smed_sulfotransferase protein* [h1SMcG0008457] (FDR = 2.05436308584779e-21, log2FC = 1.81235258125079)
2. *Smed_ankyrin repeat domain-containing protein* [h1SMcG0023020] (FDR = 2.21176621872819e-05, log2FC = 1.45058637954735)
3. *Smed_90 kDa heat shock protein* [h1SMcG0005587] (FDR = 2.27042014998624e-100, log2FC = 1.30187326090039)
4. *Smed_small HSP protein* [h1SMcG0001677] (FDR = 4.30634000225435e-23, log2FC = 1.21391746251354)
5. *Smed_small HSP protein* [h1SMcG0001680] (FDR = 6.229993549442e-44, log2FC = 1.17183167175945)
6. *Smed_small HSP protein* [h1SMcG0001675] (FDR = 7.60880192941395e-30, log2FC = 1.15856820858095)
7. *Smed_ABC transporter C family protein* [h1SMcG0013272] (FDR = 5.99707727818586e-06, log2FC = 1.15338472141863)
8. *Smed_small HSP protein* [h1SMcG0001676] (FDR = 4.9631051648721e-12, log2FC = 1.06841569053862)
9. **Smed_delta-1** [h1SMcG0016126] (FDR = 2.82694733873988e-05, log2FC = 1.03698036779469)

10. *Smed_small HSP protein* [h1SMcG0001679] (FDR=3.4876648070633e-12, log2FC=1.01777567918819)
11. *Smed_ston* [h1SMcG0006168] (FDR = 0.000826894283785631, log2FC = 1.00607109335542)
12. *Smed_ABCC/MRP-type ABC transporter* [h1SMcG0013273] (FDR = 1.99410469426834e-06, log2FC = 0.917237665130442)
13. *Smed_ABCC/MRP-type ABC transporter* [h1SMcG0013268] (FDR = 2.47708742207394e-17, log2FC = 0.911406912368186)
14. *Smed_actin-like protein* [h1SMcG0014849] (FDR = 6.8932679035396e-19, log2FC = 0.902026718983494)
15. *Smed_transposase-like protein* [h1SMnG0018121] (FDR = 0.0172608783520483, log2FC = 0.883424492333882)
16. *Smed_axonemal dynein heavy chain* [h1SMcG0022700] (FDR = 1.14884648091856e-07, log2FC = 0.874196124748799)
17. *Smed_axonemal dynein heavy chain* [h1SMcG0020686] (FDR = 0.0109580797469016, log2FC = 0.869079519562337)
18. *Smed_cytoplasmic dynein heavy chain* [h1SMcG0022860] (FDR = 2.77198901373213e-05, log2FC = 0.782067334215163)
19. *Smed_axonemal dynein heavy chain* [h1SMcG0018640] (FDR = 1.99816736988468e-07, log2FC = 0.757528546305165)
20. *Smed_Mp-2* [h1SMcG0011793] (FDR = 0.000195242034280664, log2FC = 0.752193372000123)
21. *Smed_axonemal dynein heavy chain* [h1SMcG0018055] (FDR = 0.00258730580409117, log2FC = 0.743195772749466)
22. *Smed_Ft-1* [h1SMcG0010506] (FDR = 0.000475177727251174, log2FC = 0.740874916822258)
23. *Smed_STAR-family KH domain RNA-binding protein* [h1SMcG0002667] (FDR = 0.00637663903872372, log2FC = 0.735214755651659)
24. *Smed_SOSd* [h1SMcG0011383] (FDR = 2.79980533290197e-12, log2FC = 0.715620147618047)
25. *Smed_small HSP protein* [h1SMcG0001678] (FDR=2.27205663781796e-05, log2FC= 0.70999518181106)
26. *Smed_ankyrin repeat-containing protein* [h1SMcG0019136] (FDR = 1.73059814938367e-13, log2FC = 0.706931505115801)
27. *Smed_transposon Ty3-G Gag-Pol polyprotein* [h1SMcG0017868] (FDR = 0.0053115278081507, log2FC = 0.706001097182304)
28. *Smed_nesprin-like protein* [h1SMcG0022413] (FDR = 7.02187568537665e-06, log2FC = 0.704461509131597)
29. *Smed_ABCC/MRP-type ATP-binding cassette transporter* [h1SMcG0013279] (FDR = 2.29690467822169e-09, log2FC = 0.702368667606)
30. *Smed_CCHC-type zinc finger nucleic acid-binding protein* [h1SMnG0008619] (FDR = 0.00586269755762677, log2FC = 0.700321575226876)
31. *Smed_axonemal dynein heavy chain* [h1SMcG0000295] (FDR = 2.96783561772498e-06, log2FC = 0.679575942766187)
32. *Smed_axonemal dynein heavy chain* [h1SMcG0006122] (FDR = 0.0165982717058151, log2FC = 0.664413313581688)
33. *Smed_LCAT-related protein* [h1SMcG0020471] (FDR = 0.034610987035899, log2FC = 0.653569948030889)

34. **Smed_pmp-10** [h1SMcG0013410] (FDR = 7.66154067075237e-08, log2FC = 0.648565084398496)
35. *Smed_axonemal dynein heavy chain* [h1SMcG0004574] (FDR = 7.55111169237415e-06, log2FC = 0.636459049115603)
36. *Smed_GGCT family enzyme* [h1SMcG0022107] (FDR = 7.05401295253041e-05, log2FC = 0.635483467144109)
37. **Smed_myosin heavy chain A** [h1SMcG0000998] (FDR = 1.48945159864976e-12, log2FC = 0.626508559781682)
38. *Smed_fibrinogen-related protein* [h1SMcG0017299] (FDR = 0.007115965136142, log2FC = 0.61387261303282)
39. *Smed_S-adenosyl-L-methionine-dependent methyltransferase protein* [h1SMcG0000402] (FDR = 5.78894829736525e-12, log2FC = 0.612097399809787)
40. *Smed_BLTP family protein* [h1SMcG0022922] (FDR = 0.0004588289889511, log2FC = 0.611163941424742)
41. **Smed_collagen F-10** [h1SMcG0016673] (FDR = 1.93742400769852e-05, log2FC = 0.604196028256569)
42. **Smed_slc47a-5** [h1SMcG0011309] (FDR = 5.05105096479812e-07, log2FC = 0.601223745750468)
43. *Smed_AHNAK-like large scaffold protein* [h1SMcG0001669] (FDR = 3.53636259687771e-07, log2FC = 0.600100158154358)
44. *Smed_plakin family cytoskeletal linker protein* [h1SMcG0012034] (FDR = 1.23119566401762e-11, log2FC = 0.58790245488094)
45. *Smed_ectopic P granules protein 5 homolog* [h1SMcG0003834] (FDR = 0.000405973300276857, log2FC = 0.58603498970876)

Downregulated genes at 36h (compared to earlier hypoxic time-point of 12h), excluding 'ubiquitin' hits (h1SMcG0011872 and h1SMcG0017816), ranked from higher magnitude of downregulation:

1. *Smed_cytochrome P450 2K1-like protein* [h1SMcG0010360] (FDR = 1.06989363448199e-05, log2FC = -3.58638140176185)
2. *Smed_cellular IAP repeat-containing protein* [h1SMcG0001546] (FDR = 5.34697046506412e-22, log2FC = -3.34522050141785)
3. **Smed_klf10/11** [h1SMcG0018872] (FDR = 1.06941981807827e-223, log2FC = -3.11713989988276)
4. **Smed_protein phosphatase 1 regulatory-1** [h1SMcG0010862] (FDR = 3.27926601347087e-51, log2FC = -2.1520809280067)
5. *Smed_gastric triacylglycerol lipase-like* [h1SMcG0006863] (FDR = 5.34574750734278e-11, log2FC = -1.62287441819147)
6. **Smed_histone H1** [h1SMcG0008035] (FDR = 1.31712202444632e-79, log2FC = -1.55214717220168)
7. *Smed_Kunitz-type serine protease inhibitor* [h1SMcG0000534] (FDR = 7.48477194064834e-10, log2FC = -1.48338516651355)
8. **Smed_epidermal growth factor-1** [h1SMcG0022114] (FDR = 2.44572169761692e-11, log2FC = -1.34653976833614)
9. *Smed_PTEN-like phosphatidylinositol phosphatase* [h1SMnG0004503] (FDR = 0.0179593507433655, log2FC = -1.3394471189494)

10. *Smed_histone H3-like* [h1SMcG0007872] (FDR = 0.00017450514591458, log2FC = -1.33466819157738)
11. *Smed_D-3-phosphoglycerate dehydrogenase-like* [h1SMcG0017641] (FDR = 5.30531416058592e-50, log2FC = -1.3293925611835)
12. *Smed_TNF receptor-associated factor-2-like protein A* [h1SMcG0006232] (FDR = 9.0167004846442e-28, log2FC = -1.26973666928233)
13. *Smed_hadrian* [h1SMcG0020220] (FDR = 6.83836109773274e-06, log2FC = -1.25168948157173)
14. *Smed_alkaline phosphatase-like* [h1SMcG0002552] (FDR = 2.87254126002257e-13, log2FC = -1.23191583004536)
15. *Smed_folliculin-interacting protein 2-like* [h1SMcG0006386] (FDR = 1.5014565725688e-18, log2FC = -1.22284641416131)
16. *Smed_PIM-1* [h1SMcG0003234] (FDR = 3.4678563069164e-24, log2FC = -1.22102672484891)
17. *Smed_cysteine-rich secretory protein-related CAP protein* [h1SMcG0003845] (FDR = 0.016992598323795, log2FC = -1.21718074154026)
18. *Smed_blitzschnell 3G* [h1SMcG0017123] (FDR = 0.0485514429455689, log2FC = -1.21592670812716)
19. *Smed_26S proteasome regulatory subunit ATPase* [h1SMcG0004800] (FDR = 0.0152854581872253, log2FC = -1.18725389141409)
20. *Smed_plasminogen-1* [h1SMcG0006227] (FDR = 0.00258715440885375, log2FC = -1.15165565305349)
21. *Smed_AC3-33* [h1SMcG0016352] (FDR = 0.00296600078482503, log2FC = -1.13812678445657)
22. *Smed_TNF receptor-associated factor 2* [h1SMcG0015086] (FDR = 0.00101002401666162, log2FC = -1.13451390533672)
23. *Smed_IAP domain-containing protein* [h1SMcG0007403] (FDR = 1.45540694651912e-10, log2FC = -1.08890260430628)
24. *Smed_orthopedia homeobox* [h1SMcG0022849] (FDR = 0.0114834482566607, log2FC = -1.08229114336242)
25. *Smed_ankyrin repeat-containing protein* [h1SMnG0032058] (FDR = 3.47522465368182e-09, log2FC = -1.07612269777668)
26. *Smed_E3 ubiquitin-protein ligase ARIH1* [h1SMcG0021303] (FDR = 1.03359047848803e-37, log2FC = -1.07216991011497)
27. *Smed_Plac8-E* [h1SMcG0005183] (FDR = 0.0142762094907877, log2FC = -1.02184333500747)
28. *Smed_TNF receptor-associated factor* [h1SMcG0007071] (FDR = 7.42961150185422e-13, log2FC = -1.01063170949328)
29. *Smed_GCR014* [h1SMcG0015468] (FDR = 0.0489711105063889, log2FC = -1.00886040915982)
30. *Smed_slc7a-5* [h1SMcG0021830] (FDR = 0.000602096013942355, log2FC = -0.991145459468505)
31. *Smed_60S ribosomal protein L30* [h1SMcG0003843] (FDR = 1.36385572544529e-07, log2FC = -0.985453617230265)
32. *Smed_basic-leucine zipper (bZIP) domain transcription factor* [h1SMnG0004780] (FDR = 1.34935590540759e-53, log2FC = -0.977215605314838)

33. *Smed_histone 2A-like* [h1SMcG0014269] (FDR = 0.00690560151840286, log2FC = -0.967948486681474)
34. *Smed_tissue factor pathway inhibitor-1* [h1SMnG0000861] (FDR = 0.0149210766408362, log2FC = -0.966913283750691)
35. *Smed_slc30a-2* [h1SMcG0000860] (FDR = 7.24778435203243e-06, log2FC = -0.933203469996053)
36. *Smed_Ataxin-1-like* [h1SMcG0020563] (FDR = 5.30024715507665e-09, log2FC = -0.916357736738865)
37. *Smed_TNF receptor associated factor-3* [h1SMcG0000388] (FDR = 3.71202489859942e-05, log2FC = -0.899087778277547)
38. *Smed_diacylglycerol lipase* [h1SMcG0021551] (FDR = 0.0139185598168135, log2FC = -0.885138175908692)
39. *Smed_histone H4* [h1SMnG0007240] (FDR = 0.0193996500207252, log2FC = -0.881723582969167)
40. *Smed_RNA-binding protein 48* [h1SMcG0005509] (FDR = 0.0320496102988433, log2FC = -0.867913600894462)
41. *Smed_P450 2B-like protein* [h1SMcG0009082] (FDR = 0.0207145699297744, log2FC = -0.854939591117571)
42. *Smed_tlxl* [h1SMcG0020740] (FDR = 0.00745028016026417, log2FC = -0.854254804120552)
43. *Smed_peptidoglycan recognition protein-1* [h1SMcG0017863] (FDR = 2.37121667007348e-07, log2FC = -0.850482342894666)
44. *Smed_TRAF interacting protein* [h1SMcG0018943] (FDR = 0.00836456686617417, log2FC = -0.830657351171954)
45. *Smed_tetraspanin family protein* [h1SMnG0002229] (FDR = 0.0111700314461641, log2FC = -0.830307053280063)
46. *Smed_protein O-glucosyltransferase* [h1SMcG0022134] (FDR = 0.0246444604042625, log2FC = -0.824716101372422)
47. *Smed_Mlh1* [h1SMnG0024466] (FDR = 0.002763001352862, log2FC = -0.809640548105438)
48. *Smed_monoglyceride lipase* [h1SMcG0000650] (FDR = 0.0423046005313179, log2FC = -0.801264020844012)
49. *Smed_alpha-tocopherol transfer protein-like* [h1SMcG0022482] (FDR = 0.0388835490686785, log2FC = -0.797792197508515)
50. *Smed_protein phosphatase 1 regulatory subunit* [h1SMcG0019740] (FDR = 1.15159840333954e-13, log2FC = -0.782455639859388)
51. *Smed_peptidoglycan recognition protein-1* [h1SMcG0017862] (FDR = 0.000897946643836343, log2FC = -0.767771183962991)
52. *Smed_MAPK phosphatase* [h1SMcG0019962] (FDR = 3.43071550036701e-13, log2FC = -0.764424881797494)
53. *Smed_TNF receptor-associated factor* [h1SMnG0031165] (FDR = 0.00453472409550257, log2FC = -0.761586397315326)
54. *Smed_receptor-transporting protein* [h1SMcG0006507] (FDR = 6.02894476512512e-18, log2FC = -0.761517709511887)
55. *Smed_ets-5* [h1SMcG0014005] (FDR = 0.0418649551983618, log2FC = -0.753660260618785)
56. *Smed_TNF receptor-associated protein* [h1SMcG0014828] (FDR = 9.00081126609244e-06, log2FC = -0.752689781766473)

57. **Smed_early growth response-2** [h1SMcG0000656] (FDR = 9.58565639848857e-05, log2FC = -0.752653254289842)
58. *Smed_protein phosphatase 1 regulatory subunit 3-like* [h1SMcG0004471] (FDR = 1.96941083565146e-21, log2FC = -0.749283024287614)
59. *Smed_monoglyceride lipase-like* [h1SMcG0018156] (FDR = 1.74526281930706e-09, log2FC = -0.733309939471386)
60. **Smed_fos-1** [h1SMcG0010609] (FDR = 7.62598581704811e-23, log2FC = -0.728127005000873)
61. *Smed_steroid 17-alpha-hydroxylase/17,20 lyase-like* [h1SMcG0005150] (FDR = 8.62204157837406e-16, log2FC = -0.725318794437626)
62. **Smed_1-Jun** [h1SMcG0021300] (FDR = 3.52377490378272e-13, log2FC = -0.712648715798711)
63. *Smed_methylenetetrahydrofolate reductase (NADPH)-like* [h1SMcG0005986] (FDR = 8.02681260794493e-13, log2FC = -0.711300469958619)
64. *Smed_serine/threonine-protein kinase* [h1SMcG0017420] (FDR = 0.00972170350340855, log2FC = -0.703370702484254)
65. *Smed_JNK1/MAPK8-associated membrane protein-like* [h1SMcG0003989] (FDR = 3.72149762361601e-06, log2FC = -0.695018367078084)
66. *Smed_lipopolysaccharide-induced tumour necrosis factor-alpha factor* [h1SMcG0015134] (FDR = 1.83977795671166e-05, log2FC = -0.694401807934568)
67. *Smed_ionotropic glutamate receptor (ligand-gated ion channel)* [h1SMcG0012789] (FDR = 0.0472337669496643, log2FC = -0.686673649481756)
68. *Smed_MFS transporter* [h1SMcG0021070] (FDR = 0.000327029068617646, log2FC = -0.685987876156948)
69. *Smed_strictosidine synthase-like* [h1SMcG0015357] (FDR = 0.0104886685171892, log2FC = -0.679856768685133)
70. *Smed_phytanoyl-CoA dioxygenase domain-containing protein* [h1SMcG0021982] (FDR = 0.017327103381154, log2FC = -0.677922081071219)
71. *Smed_synaptotagmin family protein* [h1SMcG0005640] (FDR = 0.0464315863461809, log2FC = -0.662111112281362)
72. *Smed_Hsp40 co-chaperone, subfamily C* [h1SMcG0002360] (FDR = 0.0272260556064815, log2FC = -0.654102674404845)
73. *Smed_VWFC domain-containing protein* [h1SMcG0004289] (FDR = 0.000734690466917389, log2FC = -0.651667313126719)
74. *Smed_cramped-like* [h1SMcG0002801] (FDR = 0.0118945774774986, log2FC = -0.650104431855401)
75. *Smed_axonemal dynein light chain* [h1SMnG0013155] (FDR = 0.0219625387457349, log2FC = -0.636101809310618)
76. **Smed_slc25a-29** [h1SMcG0020425] (FDR = 7.27546114508417e-13, log2FC = -0.635154285159658)
77. **Smed_GCR083** [h1SMcG0003850] (FDR = 0.0231924321272808, log2FC = -0.633335786230609)
78. **Smed_slc22a-2** [h1SMcG0008448] (FDR = 0.00060543566529015, log2FC = -0.633077297055207)
79. *Smed_cyclin protein* [h1SMcG0022222] (FDR = 1.12423740930592e-11, log2FC = -0.631967620193513)
80. *Smed_E3 ubiquitin ligase RBR family protein* [h1SMcG0004408] (FDR = 1.53575366623451e-13, log2FC = -0.631337799756321)

81. *Smed_cytochrome P450 superfamily protein* [h1SMcG0010229] (FDR = 0.00850518366600257, log2FC = -0.630415955453325)
82. *Smed_TEX261* [h1SMcG0002340] (FDR = 0.00421304877511844, log2FC = -0.627330079227369)
83. *Smed_X1.A4.3* [h1SMnG0020921] (FDR = 0.000114196744823142, log2FC = -0.624840559092585)
84. *Smed_IAP domain containing-1* [h1SMcG0007420] (FDR = 0.000238515517051844, log2FC = -0.619915248492393)
85. *Smed_EGR-like protein 1* [h1SMcG0008212] (FDR = 0.00279186509750272, log2FC = -0.61825305839441)
86. *Smed_family A GPCR* [h1SMcG0012991] (FDR = 0.041124009757083, log2FC = -0.612766497331541)
87. *Smed_chloride intracellular channel protein* [h1SMcG0012197] (FDR = 0.0440458185571227, log2FC = -0.610564230275572)
88. *Smed_rhodanese-like domain protein* [h1SMcG0021461] (FDR = 1.0163574242913e-11, log2FC = -0.602458383696163)
89. *Smed_sideroflexin-1* [h1SMcG0006070] (FDR = 1.03922435001556e-08, log2FC = -0.601794256519659)
90. *Smed_phospholipase A2-like* [h1SMcG0017624] (FDR = 4.77402337005204e-10, log2FC = -0.595434300929813)
91. *Smed_Rab family small GTPase* [h1SMcG0007171] (FDR = 0.0308948838804529, log2FC = -0.592959000789082)

24h vs 36h (FC = 1.5, FDR 0.05)

Upregulated genes at 36h (compared to earlier hypoxic time-point of 24h) ranked from higher magnitude of upregulation:

1. *Smed_delta-1* [h1SMcG0016126] (FDR = 5.00992508511947e-06, log2FC = 1.41990414509384)
2. *Smed_ston* [h1SMcG0006168] (FDR = 2.62086910122206e-05, log2FC = 1.3869222151307)
3. *Smed_90 kDa heat shock protein* [h1SMcG0005587] (FDR = 7.6550536926258e-27, log2FC = 1.07081726015625)
4. *Smed_diamine acetyltransferase-like* [h1SMcG0002311] (FDR = 0.00013207917996415, log2FC = 0.798000714658877)
5. *Smed_sulfotransferase protein* [h1SMcG0008457] (FDR = 0.00320820491215593, log2FC = 0.78614790006495)
6. *Smed_small HSP protein* [h1SMcG0001680] (FDR = 0.000135009735318466, log2FC = 0.667553270346919)
7. *Smed_small HSP protein* [h1SMcG0001677] (FDR = 0.000288140738962214, log2FC = -0.66647954533951)
8. *Smed_actin-like protein* [h1SMcG0014849] (FDR = 0.000248223927640637, log2FC = 0.62829693980447)
9. *Smed_small HSP protein* [h1SMcG0001675] (FDR = 0.000500805877905013, log2FC = 0.617994641265033)

Downregulated genes at 36h (compared to earlier hypoxic time-point of 24h) ranked from higher magnitude of downregulation, excluding 'ubiquitin' and 'polyubiquitin' hits (h1SMcG0020974, h1SMcG0020971, h1SMcG0020969, h1SMcG0020966, h1SMcG0020967, h1SMcG0020968, h1SMcG0020970, h1SMcG0020973 and h1SMcG0020975):

1. *Smed_cellular IAP repeat-containing protein* [h1SMcG0001546] (FDR = 2.55344269766299e-30, log2FC = -5.01543901340017)
2. *Smed_cytochrome P450 2K1-like protein* [h1SMcG0010360] (FDR = 1.01628047593439e-05, log2FC = -3.16927654423405)
3. *Smed_klf10/11* [h1SMcG0018872] (FDR = 2.71864977014386e-213, log2FC = -2.98862376963382)
4. *Smed_TNF receptor-associated factor 2* [h1SMcG0015086] (FDR = 9.10335736425316e-19, log2FC = -2.58314792264711)
5. *Smed_E3 ubiquitin-protein ligase ARIH1* [h1SMcG0021303] (FDR = 4.62628524681727e-103, log2FC = -2.19941209287836)
6. *Smed_epidermal growth factor-1* [h1SMcG0022114] (FDR = 1.7095309242955e-16, log2FC = -1.96519070945644)
7. *Smed_E3 ubiquitin ligase RBR family protein* [h1SMcG0004408] (FDR = 1.30560088371346e-127, log2FC = -1.90741894045977)
8. *Smed_pantothenate kinase protein* [h1SMcG0016781] (FDR = 1.11876396081675e-33, log2FC = -1.6588550261494)
9. *Smed_histone H1* [h1SMcG0008035] (FDR = 4.2229878895499e-31, log2FC = -1.50102739981944)
10. *Smed_TNF receptor-associated factor* [h1SMcG0007071] (FDR = 7.65324581966775e-22, log2FC = -1.46374192806526)
11. *Smed_soma ferritin* [h1SMcG0007370] (FDR = 1.62457841553092e-30, log2FC = -1.44197621099769)
12. *Smed_gastric triacylglycerol lipase-like* [h1SMcG0006863] (FDR = 1.8548203804286e-05, log2FC = -1.43801344389435)
13. *Smed_TNF receptor-associated factor-2-like protein A* [h1SMcG0006232] (FDR = 5.51331277585415e-21, log2FC = -1.38876631709852)
14. *Smed_protein phosphatase 1 regulatory-1* [h1SMcG0010862] (FDR = 4.90506593231001e-14, log2FC = -1.36888968730558)
15. *Smed_family A G protein-coupled receptor* [h1SMcG0009373] (FDR = 0.00141691017138097, log2FC = -1.3313639998243)
16. *Smed_TNF receptor-associated factor* [h1SMcG0022911] (FDR = 0.00012771071382944, log2FC = -1.32206135694797)
17. *Smed_IAP domain-containing protein* [h1SMcG0007403] (FDR = 0.00160736942302202, log2FC = -1.316357692833)
18. *Smed_MAPK phosphatase* [h1SMcG0019962] (FDR = 2.25559806095537e-31, log2FC = -1.27774734739345)
19. *Smed_basic-leucine zipper (bZIP) domain transcription factor* [h1SMnG0004780] (FDR = 1.50255107950974e-12, log2FC = -1.25967265093886)
20. *Smed_histone H3-like* [h1SMcG0007872] (FDR = 0.0132317632248617, log2FC = -1.25674144087214)

21. *Smed_pantothenate kinase protein* [h1SMcG0016782] (FDR = 3.08199077196035e-22, log2FC = -1.23223973263922)
22. *Smed_D-3-phosphoglycerate dehydrogenase-like* [h1SMcG0017641] (FDR = 4.67055133799836e-23, log2FC = -1.20409620033173)
23. *Smed_PIM-1* [h1SMcG0003234] (FDR = 3.58854209991857e-05, log2FC = -1.18277737488765)
24. *Smed_RNA-binding protein 48* [h1SMcG0005509] (FDR = 0.0231962587697595, log2FC = -1.11611361035756)
25. *Smed_monocarboxylate transporter protein* [h1SMcG0016325] (FDR = 0.000457514030902915, log2FC = -1.08160823834942)
26. *Smed_consortin-like* [h1SMcG0022227] (FDR = 0.0448971373435474, log2FC = -1.06265448451436)
27. *Smed_receptor-transporting protein* [h1SMcG0006507] (FDR = 8.58666599099691e-11, log2FC = -1.0578799633052)
28. *Smed_ferritin* [h1SMcG0007372] (FDR = 1.11194516403082e-15, log2FC = -1.0461683658943)
29. *Smed_peptide methionine sulfoxide reductase* [h1SMcG0020033] (FDR = 1.80260603354743e-09, log2FC = -1.02823849246589)
30. *Smed_monoglyceride lipase-like* [h1SMcG0018156] (FDR = 3.57208299761192e-09, log2FC = -1.02580959447097)
31. *Smed_Kunitz-type serine protease inhibitor* [h1SMcG0000534] (FDR = 0.0266009393564645, log2FC = -1.02175295870548)
32. *Smed_dynamin-1-like* [h1SMcG0013705] (FDR = 4.64122106031773e-13, log2FC = -1.0046794193863)
33. *Smed_protein phosphatase 1 regulatory subunit* [h1SMcG0019740] (FDR = 2.1935650381622e-20, log2FC = -0.997677061697948)
34. *Smed_IAP domain containing-1* [h1SMcG0007420] (FDR = 2.87681337122496e-10, log2FC = -0.996460509257607)
35. *Smed_blitzschnell 3G* [h1SMcG0017123] (FDR = 0.00762798014537809, log2FC = -0.989628408774458)
36. *Smed_cysteine-rich secretory protein-related CAP protein* [h1SMcG0003845] (FDR = 0.00940340313180608, log2FC = -0.981770614425919)
37. *Smed_slc24a-5* [h1SMnG0034353] (FDR = 6.39514600670735e-05, log2FC = -0.95033046806843)
38. *Smed_slc20a-2* [h1SMcG0001112] (FDR = 0.000119762002940969, log2FC = -0.947909113131601)
39. *Smed_1-Jun* [h1SMcG0021300] (FDR = 2.04136210621929e-14, log2FC = -0.932212875472068)
40. *Smed_serine/threonine-protein kinase* [h1SMcG0017420] (FDR = 4.58100187720049e-07, log2FC = -0.925663427291129)
41. *Smed_protein phosphatase 1 regulatory subunit 3-like* [h1SMcG0004471] (FDR = 1.19015064136931e-20, log2FC = -0.920334924330868)
42. *Smed_tlx* [h1SMcG0020740] (FDR = 0.0368704786913544, log2FC = -0.91157455695737)
43. *Smed_dual serine/threonine and tyrosine protein kinase-like* [h1SMcG0012240] (FDR = 7.07817796236198e-10, log2FC = -0.911150492011115)
44. *Smed_folliculin-interacting protein 2-like* [h1SMcG0006386] (FDR = 6.68838950209984e-06, log2FC = -0.89925972343877)

45. **Smed_peptidoglycan recognition protein-1** [h1SMcG0017862] (FDR = 1.78139464960771e-05, log2FC = -0.899210438527305)
46. *Smed_5'-AMP-activated protein kinase, gamma regulatory subunit* [h1SMcG0005059] (FDR = 2.1935650381622e-20, log2FC = -0.891966514896042)
47. *Smed_EF-hand domain-containing protein* [h1SMcG0009513] (FDR = 1.316186684528343-17, log2FC = -0.889907403363789)
48. *Smed_ABHD superfamily protein* [h1SMcG0011435] (FDR = 3.55163481433788e-09, log2FC = -0.884237125443796)
49. *Smed_phospholipase A2-like* [h1SMcG0017624] (FDR = 1.99748229148943e-11, log2FC = -0.860258785756353)
50. *Smed_universal stress protein* [h1SMcG0009643] (FDR = 1.83852916983496e-12, log2FC = -0.83764931209418)
51. **Smed_slc25a-29** [h1SMcG0020425] (FDR = 2.77686970814052e-15, log2FC = -0.828131589696375)
52. *Smed_metallophosphoesterase* [h1SMcG0000917] (FDR = 1.88599369822864e-09, log2FC = -0.817927575524605)
53. **Smed_peptidoglycan recognition protein-1** [h1SMcG0017863] (FDR = 0.0144482456391352, log2FC = -0.802114701301478)
54. *Smed_cramped-like* [h1SMcG0002801] (FDR = 0.0168744507628867, log2FC = -0.801610238331739)
55. *Smed_C2H2-type zinc finger domain-containing protein* [h1SMcG0000013] (FDR = 9.7981835413962e-07, log2FC = -0.797382275956501)
56. *Smed_globin-like superfamily protein* [h1SMcG0015726] (FDR = 0.0366709272692888, log2FC = -0.789381331715145)
57. *Smed_adiponectin receptor protein-like* [h1SMcG0013451] (FDR = 6.56297899282113e-19, log2FC = -0.768521882175271)
58. **Smed_lhx2/9-1** [h1SMcG0003155] (FDR = 0.0232123776118597, log2FC = -0.765576437944006)
59. **Smed_TNF receptor associated factor-3** [h1SMcG0000388] (FDR = 0.0245010374854757, log2FC = -0.760782663712072)
60. *Smed_methylenetetrahydrofolate reductase (NADPH)-like* [h1SMcG0005986] (FDR = 2.10435374099238e-07, log2FC = -0.758404246663364)
61. *Smed_FERM domain-containing protein* [h1SMcG0005416] (FDR = 2.64379364471324e-05, log2FC = -0.734704406031016)
62. **Smed_caspase-3C** [h1SMcG0020890] (FDR = 0.00141691017138097, log2FC = -0.729294252680509)
63. *Smed_LRR-containing protein* [h1SMcG0015004] (FDR = 1.80260603354743e-09, log2FC = -0.692636904607545)
64. *Smed_cytochrome P450 superfamily protein* [h1SMcG0006631] (FDR = 1.25397496720097e-11, log2FC = -0.689256542365537)
65. *Smed_pseudo-PARP domain-containing protein* [h1SMcG0001523] (FDR = 1.99748229148943e-11, log2FC = -0.683405327484958)
66. *Smed_amine oxidase* [h1SMcG0009368] (FDR = 2.68594045016764e-06, log2FC = -0.68320892742675)
67. *Smed_Ras-like guanine nucleotide exchange factor* [h1SMcG0007247] (FDR = 5.7442374025852e-06, log2FC = -0.658850885516676)

68. *Smed_single-stranded DNA-binding protein* [h1SMcG0007004] (FDR = 0.000365243198945197, log2FC = -0.657227496649509)
69. *Smed_JmjC domain-containing lysine-specific demethylase* [h1SMcG0011597] (FDR = 0.000295630874200159, log2FC = -0.657057068961271)
70. *Smed_growth factor receptor-bound protein* [h1SMcG0006330] (FDR = 0.00306457063327834, log2FC = -0.652618322967573)
71. *Smed_Ras-family small GTPase* [h1SMcG0005592] (FDR = 2.92117024081199e-07, log2FC = -0.637018348810809)
72. *Smed_elf3* [h1SMcG0016104] (FDR = 0.000262931249945659, log2FC = -0.636512276365537)
73. *Smed_steroid 17-alpha-hydroxylase/17,20 lyase-like* [h1SMcG0005150] (FDR = 5.7442374025852e-06, log2FC = -0.624896314552459)
74. *Smed_TAO family serine/threonine protein kinase* [h1SMcG0005943] (FDR = 0.00580432097791807, log2FC = -0.623740307058481)
75. *Smed_DYRK family dual-specificity protein kinase* [h1SMcG0001989] (FDR = 3.73052177683319e-06, log2FC = -0.618872751244139)
76. *Smed_2-aminoethanethiol dioxygenase* [h1SMcG0005435] (FDR = 1.54257361274794e-05, log2FC = -0.617135342548496)
77. *Smed_phosphoserine aminotransferase-like* [h1SMcG0020665] (FDR = 1.53631946261257e-10, log2FC = -0.611068863078496)
78. *Smed_TNF receptor-associated protein* [h1SMcG0014828] (FDR = 0.0232123776118597, log2FC = -0.606344727819287)
79. *Smed_mpv17-like protein* [h1SMcG0021924] (FDR = 0.0049284783613834, log2FC = -0.603660760236883)
80. *Smed_ATP-dependent RNA helicase* [h1SMcG0014297] (FDR = 9.37719592177489e-09, log2FC = -0.595822913627132)
81. *Smed_early growth response-2* [h1SMcG0000656] (FDR = 0.0289361879224237, log2FC = -0.594396698347975)
82. *Smed_Ras-family small GTPase* [h1SMcG0015679] (FDR = 0.0151880864092741, log2FC = -0.590318120259328)